Differential Dynamic Causal Nets: Model Construction, Identification and Group Comparisons

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Abstract

Pathophysiological modelling of brain systems from microscale to macroscale remains difficult in group comparisons partly because of the infeasibility of modelling the interactions of thousands of neurons at the scales involved. Here, to address the challenge, we present a novel approach to construct differential causal networks directly from grouped electroencephalogram (EEG) data. In the proposed modular network, each node represents a locally averaged regional neural system via a canonical Jansen-Rit model while directed edges stand for node-wise input-output connections. The network is hierarchically structured in the sense that node and edge parameters are random, varying in subjects, but follow a mixed-effects model. We show that unlike the classical Jansen-Rit models, the proposed node models are locally structurally identifiable. An evolutionary optimisation algorithm for parameter inference in the proposed method is developed using a loss function derived from Chen-Fliess expansions of stochastic differential equations. The method is demonstrated by fitting the proposed modular network to real grouped epileptic EEG data in both sensor- and source- levels. We identify significant changes in the parameters that characterise dynamic networks within and between brain regions. We reveal network disruptions, due to imbalance of excitatory-inhibitory interneurons and other altered epileptic brain connectivities, before and during seizure periods, and contrast them between case and control groups. The performance of the proposed method is also evaluated on synthetic data.

Key words: Differential network of dynamic causal models, M/EEG scans, epileptic brain network, dynamic causal nets, and mixed-effects models.

1 Introduction

Non-invasive imaging techniques electro- and magneto-encephalography (EEG and MEG) are often used for dynamic causal discovery and causality quantification in group comparison with neurological disorders such as stroke, motor neuron disease, Alzheimer's disease, Parkinson's disease, and epilepsy (Choi et al., 2023; Aljalal et al., 2022; Ghassemkhani et al., 2025; Pyrzowski et al., 2015; Ogrim et al., 2012; Rosch et al., 2018) among others. M/EEG are promising tools for studying brain dynamics and enable researchers and clinicians to measure and characterise brain activity in fine

temporal detail thus improving the understanding of the etiology of neurological conditions as well as allowing the development of brain-based technologies.

M/EEG imaging revealed a rich catalogue of behaviour in both resting and active brains.. For example, Papadopoulou et al. (2015) used spectral EEG data to profile epileptic seizure activity in the brain. Özbek et al. (2015) demonstrated that the frontal alpha/theta power ratio derived from the resting-state EEG data is a promising biomarker for distinguishing early-onset Alzheimer's disease from healthy controls. Duffy and Als (2012) found that EEG spectral coherence discriminates children with autism from healthy controls.

Using MEG spectral data, van Es et al. (2025) showed that, although brain network dynamics are stochastic, their activities have a robust cyclical pattern. Zhang and Green (2025) proposed a novel method for detecting mild traumatic brain injury by using spectral likelihood ratio statistics of MEG scan data when controls are not homogenous. Coronel-Oliveros et al. (2026) characterised a previously introduced multiple frequency Jansen-Rit neural mass model with inhibitory synaptic plasticity.

Complementary to the aforementioned spectral analysis, dynamic causal modelling (DCM) of M/EEG data offers a powerful way to infer effective connectivity within the brain through coupled neuronal circuits with certain biophysics-informed parameters that are responsible for what are observed in the brain (Friston et al., 2003; Kiebel et al., 2008; Grefkes et al., 2008, 2010; Herz et al., 2014; Rehme et al., 2011; Rosch et al., 2018; Forrester et al., 2020; Singh et al., 2025). Effective connectivity, referring to the causal influences that different parts of the brain exert over each other, provides a direct understanding of the neural mechanisms underlying various cognitive processes in the brain. Instead of modelling individual neurons, DCM describes the average behaviour of neural populations. In particular, allowing for dynamic causal interactions between distinct brain regions, DCM can reveal biophysics-informed links between microscopic neural activity and macroscopic brain phenomena. By adjusting model parameters, researchers can simulate the transition from normal brain rhythms to pathological states, providing valuable insights into disease mechanisms and potential treatment strategies (Wendling et al., 2024; Friston et al., 2019). Sotero et al. (2007) used neural circuits to simulate EEG signals, helping to elucidate how the macroscopic electrical activity recorded from the scalp emerges from underlying neural dynamics. Using the DCM approach, Papadopoulou et al. (2015) tracked slow modulations in synaptic gain for a cohort of epileptic patients. Chen et al. (2009) employed spectral DCM on MEG scans to infer functional asymmetries in inter- and cross-frequency-coupling between brain regions during the period that subjects visually processes faces. This approach can be used to model brain activity on a whole brain scale. With the Human Connectome Project dataset, Singh et al. (2025) developed a large scale, data-driven mean field cortical model for activities in the brain, involving hundreds of interacting neural populations.

Following Friston et al. (2019), a multiple cortical column DCM can be described by the following set of neural mass stochastic differential equations:

$$d\boldsymbol{x}(t) \quad = \quad f(\boldsymbol{x}(t), \boldsymbol{u}(t); \boldsymbol{\theta}, \boldsymbol{K})dt + \Gamma(\boldsymbol{\theta})d\boldsymbol{w}(t), \qquad \boldsymbol{x}(0) = \boldsymbol{x}_0; \qquad y(t) = h(\boldsymbol{x}(t)),$$

where $\boldsymbol{w}(t)$ is a multivariate Gaussian white noise process, $\boldsymbol{x}(t)$ is a vector of p hidden states, $\boldsymbol{x}_0$ contains initial values of states, $y(t)$ is the partial observation of hidden states, $\boldsymbol{u}(t)$ is an external input, f is a nonlinear vector field, h is a function from $\mathbb{R}^p \to \mathbb{R}$ with $\boldsymbol{\theta}$ being a vector of local node parameters, $\boldsymbol{K}$ a matrix of connection parameters between nodes, $\Gamma(\boldsymbol{\theta})$ is a degenerate diffusion coefficient matrix. In the past two decades, considerable progress has been made in the development of dynamic causal modelling (Friston et al., 2019; Park et al., 2021; Forrester et al., 2020; Singh et al., 2025), and references therein. Despite this, a brain-wide search for dynamic

causal abnormalities across spatial scales remains difficult, partly because of the challenges in modelling neural connectivity at a large scale.

The first challenge is about how to solve and sample from high-dimensional nonlinear neural mass differential equations when the parameters involved are only partially identifiable. Kulik et al. (2023) used a fourth order stochastic Runge-Kutta (RK) method to numerically solve the coupled differential equations involved in the Jansen-Rit model. In these situations, recomputing the Jacobian-updating integrator continuously is required to solve the equations, making the computation infeasible when fitting many pairwise coupled DCMs to the data. On the other hand, a truncated Chen-Fliess expansion (Chen, 1954; Fliess, 1981) can solve the above differential equations. However, it still remains difficult to perform a Bayesian data analysis when the vector fields are highly nonlinear.

The second challenge is that some states $\boldsymbol{x}(t)$ are only partially observed at discrete time points. Then the exact likelihoods for the transition densities are difficult to trace. To estimate these likelihoods, Pokern et al. (2009) proposed a data-augmentation algorithm to impute the hidden states. The computation was intensive and it did not always converge to the true values when many local hypoelliptic diffusions were included.

The third challenge is about how to implement large scale dynamic causal models, subject to a limited resource of computation (e.g., Honey et al., 2007; Li et al., 2020; Rosch et al., 2018; Singh et al., 2025). To reduce the model dimension, Forrester et al. (2020) made a very restrictive assumption that the cortical columns are homogeneous in the sense that the column parameters are kept the same across the columns. Ziaeemehr et al. (2025) presented a valuable software package, named "Virtual Brain Inference" (VBI), that enables faster and more efficient inference of parameters in dynamical system models of whole-brain activity.

The fourth challenge is about heterogeneity of human brains even when case and control subjects are matched by ages and genders (Zhang and Green, 2025; Li et al., 2025). It is unclear how to take individual brain differences into account in a dynamic causal model with M/EEG data. Finally, although the excitation-inhibition imbalance across columns has been shown to underlie an epileptic form of brain activity (e.g., Friston et al., 2016), it is unclear how such a spatial distribution pattern is associated with epilepsy.

Here, to address these challenges, we first develop an EEG data-driven network of conditionally coupled dynamic causal models (NccDCM) across multiple cortical regions for cases and controls respectively, where each node corresponds to a functional unit that can be represented by a neural mass model, and with edges informed by effective connectivity. For simplicity, neural mass activities are described by stochastic Jansen-Rit models (Jansen and Rit, 1995; Ableidinger et al., 2017; Forrester et al., 2020).

We then calculate differential causal nets between cases and controls to quantify abnormal causality in patients. By these nets, we are able to identify intrinsic and extrinsic dynamic connectivity patterns that are associated with brain diseases. In the proposed network, each node describes a channel of neuron populations which may receive input signals from other channels. The strengths of directed connectivity between nodes are regularised by biophysics-informed parameters.

The above network structure allows sub-networks to be nested within a composite network, thereby providing a basis for the subsequent differential analysis of brain systems at different levels of detail. The following assumptions are made in the NccDCM: Within each channel/node, the neurons can be modelled by three populations, namely, pyramidal cells (P), excitatory interneurons (E) and inhibitory interneurons (I), forming two feedback loops- a fast excitatory feedback loop with

external input from paired opposite regions as well as an input of rough path noise from background; a slow inhibitory feedback loop. The output represents the membrane potential of pyramidal neurons, which are signals measured by the EEG device. The coupling within a node and between nodes are regularised by structural parameters. However, it should be noted that the precise nature of the within channel model is not restrictive and more complex models can be incorporated.

A conditional inference of the proposed model is implemented as follows: We first estimate NccDCMs by ccDCM screening, followed by inferring disease-associated causal nets from estimated contrasts between case and control subjects. The conditional estimation is carried out through minimising a profiled loss function based on what is called Chen-Fliess expansions of stochastic differential equations (Ableidinger et al., 2017; Chen, 1954; Fliess, 1981; Lyons, 1998). In the Chen-Fliess expansions, we solve the stochastic differential equations by linear combinations of iterative integrals of the vector of field functions. These estimated ccDCMs are then assembled via a post analysis of variance (Anderson, 2017). Unlike the existing approaches, which assume parameters to be known from diffusion-imaging data and previous literature (e.g., Forrester et al., 2020), we directly estimate local parameters in each node conditionally on inputs from other nodes by using subject's EEG data. This enables the inference of the large number of node parameters feasible even when nodes are heterogeneous. The estimated NccDCM allows us to investigate how neural activities in these channels/nodes influence each other, particularly, when defected channels/nodes exhibit pathological dynamics compared to healthy controls. Changes in these parameters reflect alterations in the balance between excitation and inhibition, compensatory adaption to abnormal inputs. We evaluate the proposed methodology on both simulated data and a recently published benchmark dataset of scalp EEG recordings for a group of pediatric epilepsy patients and their controls (Fasil et al., 2021). See Figure S4, the Online Supplementary Materials for an example of interictal epileptiform discharges (IEDs) in the data and the bifurcation behaviour of the fitted ccDCM.

A healthy, balanced brain connectivity is critical for maintaining brain functions. In contrast, an abnormal brain connectivity can change subject's behavior by affecting certain areas involved in seizure propagation. Epilepsy is a group of neurological disorders characterised by abnormal spontaneous brain activity, involving multiscale changes in brain functional organizations. The system epilepsy hypothesis posits that the enduring susceptibility to generate seizures in epilepsy is due to the specific vulnerability of the brain system (Fisher and et al., 2005). Abnormal firing of neurons causes defected local circuit function, leading to alterations of macroscale functional activities during seizure propagation (Burman and Parrish, 2018). The virtual brain model has demonstrated that a combination of a global shift in the brain's dynamic equilibrium and locally hyperexcitable network nodes provides a mechanistic explanation for the epileptic brain during interictal resting state (Courtiol et al., 2020; Yang et al., 2024). IED in routine EEG recordings is crucial evidence of epilepsy in patients. Fasil et al. (2021) published a benchmark dataset for detecting IED. However, it is not clear to what extent the IEDs affect macroscale intrinsic dynamics and microcircuit organizations, that supports their pathological relevance. In the real epileptic data analysis, we treat IEDs as electroencephalographic markers for epilepsy. IEDs reflect transient cortical hyperexcitability in neuron populations and are useful biomarkers in diagnosis and treatment of epilepsy. We are in particular interested in characterising the IED associated epileptic networks via differential dynamic causal nets, which shed light on the mechanisms of seizure propagation in brain regions.

The contributions of this paper are two-fold. First, we develop a novel network architecture that leverage the region-wise coupling across a whole brain via

hierarchical mixed-effects modelling. We provide a conditional inference scheme to estimate NccDCM. By the post analysis of variance, we demonstrate that the proposed modelling is promising in discovering novel differential dynamic causal nets and in providing a deep insight into abnormal connectivity in case subjects compared to healthy controls. Secondly, fitting the proposed NccDCM to an epileptic case-control dataset in sensor and source levels, we identify comprehensive neuronal circuits for a whole brain and reveal biophysics-informed dynamic causal nets that underlie defected brain connectivity in epileptic patients.

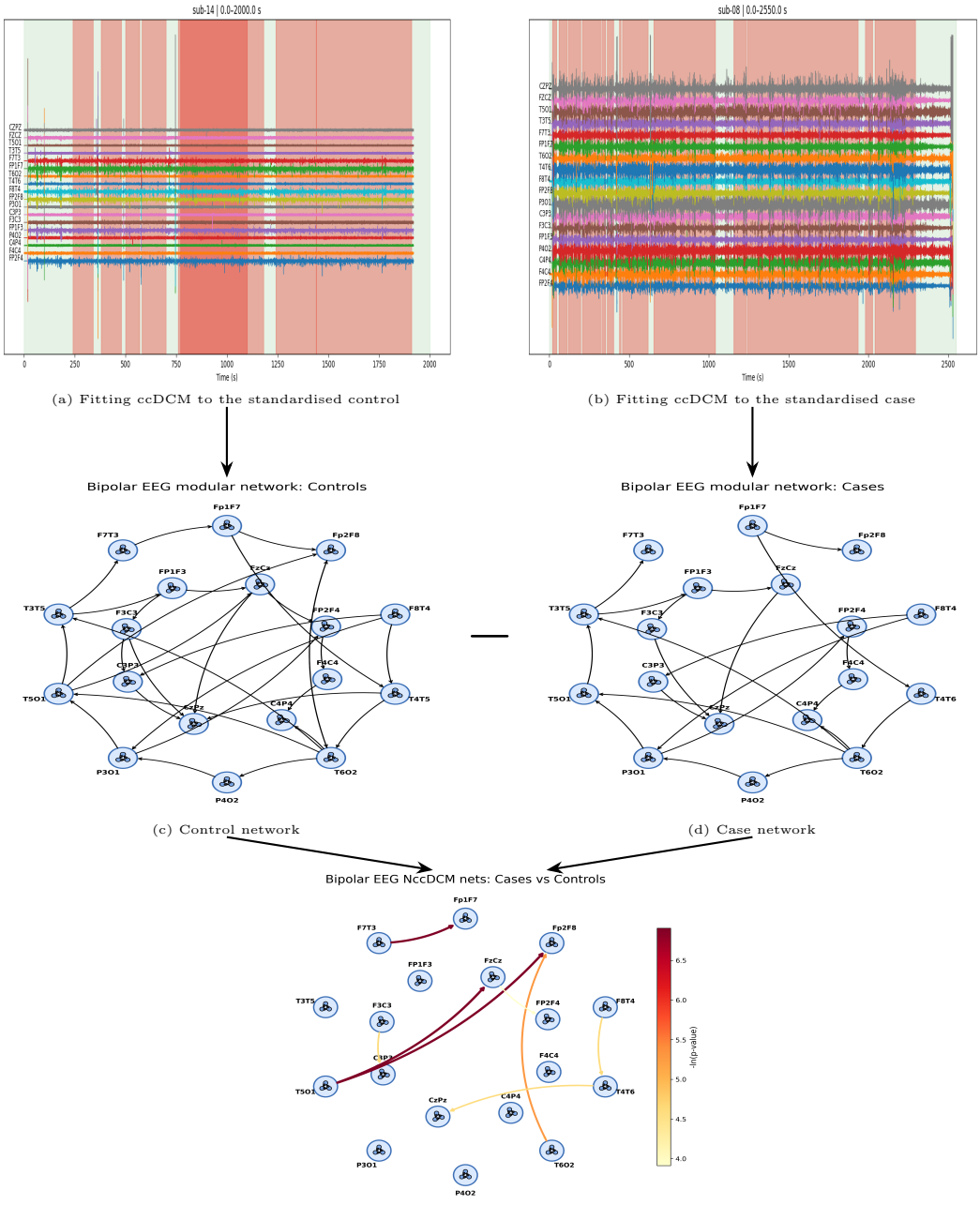
The remaining paper is organised as follows. The details of the proposed methodology and optimisation algorithm are provided in Section 2. The applications of the proposed methods to the IED dataset and synthetic data are presented in Section 3. Discussion and conclusions are made in Section 4.

2 Methods and materials

In this section, we model subject's EEG data using modular networks, where each node is represented by a conditionally coupled dynamic causal model (ccDCM). By conditional coupling, we mean that the output of each node is treated, in turn, as the response variable, while the neural populations within that node serve as internal regressors and the other nodes serve as external regressors. We further assume that the model parameters for subjects in the groups follow a mixed-effects model, thereby accounting for both group-level and subject-specific variations. The ccDCM for each node is then estimated by minimising an estimated loss function, defined as minus the estimated log-likelihood, using an evolutionary optimisation algorithm. The loss function is derived from the Chen–Fliess solution to the stochastic Jansen–Rit differential equations. Finally, edges in differential dynamic causal nets are identified by testing for significant differences in the model parameters between the groups using a post hoc permutational multivariate analysis of variance (PERMANOVA) (Anderson, 2017).

2.1 Models

To account for multiple nested sources of variability in the data, the network of ccDCMs (NccDCM) is constructed hierarchically, with observable outputs from each node conditionally on certain parameters, which themselves are given a probabilistic specification. Nodes connect each other via directed arrows with transmission rates. The node and edge parameters are modelled using a mixed-effects model to capture both group-level and subject-specific variations. To identify NccDCM differences between the groups, for each pair of nodes, treating one as an output node and the other as an input node, we test for differences between two NccDCMs per parameter. We assign a directed edge from the input node to the output node if the testing result is significant after multiple testing corrections. This gives rise to a differential DCM net per parameter. Their performance can deteriorate by confounding effects of indirect and direct inputs. For example, suppose that the data are produced from a three-node ccDCM model where node 1 links to node 2 while node 2 links node 3. Given the data, a false direct link from node 1 to node 3 can be generated from the pairwise comparison. This problem can be addressed by inputting both nodes 1 and 2 into the ccDCM. Under the multiple input framework, the influence of node 1 on node 3 is often weaker than the influence of node 2 on node 3 in terms of p-values. Therefore, the edge importances, quantified in terms of p-values, can be used to distinguish between direct and indirect links between nodes. In practice, for each node, we first select marginally significant inputs as multiple inputs in the model, refit the



(a) Fitting ccDCM to the standardised control

(b) Fitting ccDCM to the standardised case

(c) Control network

(d) Case network

(e) Differential net

Figure 1. Flowchart for generating differential NccDCM nets from sensor data. Step 1: Case and control time series are standardised and segmented (first row), and the corresponding modular networks are derived. Red shading indicates ictal periods. Each network node comprises three interconnected subcircles representing the three-state Jansen–Rit model. Step 2: Group of control modular networks are subtracted from the corresponding group of case networks via their parameters (second row). Step 3: Statistical significance testing of the network differences yields differential NccDCM nets. Step 4: Adjustment for multiple input effects. The colour bars are in the scale of $-\ln(\text{p-value})$, where p-value is BH-adjusted.

model to the data and keep those who can survive in the Benjamini-Hochberg multiple testing adjustment. Such an adjustment is called multiple input adjustment. The architectures of NccDCM and differential DCM nets are illustrated by the schematic diagrams in Figure 1. The further mathematical details are given as follows.

NccDCM and its identifiability. Each node is represented by the Jansen-Rit model, a biologically inspired model of the three interconnected neural populations in a cortical column: pyramidal projection neurons, excitatory interneurons, and inhibitory interneurons. These populations form feedback loops, enabling the model to capture the complex dynamics of cortical activity. At dendritic spines, incoming spikes are converted into postsynaptic potentials (PSPs) through convolution with an alpha-type response function. Let hidden states x_0, x_1 and x_2 denote the hidden states of these populations in terms of total membrane potential alterations in the synapses respectively. To simplify the notations, we also let L_a and L_b denote the differential operators,

$$\begin{aligned} L_a &= \partial^2/\partial t^2 + 2a\partial/\partial t + a^2, \\ L_b &= \partial^2/\partial t^2 + 2b\partial/\partial t + b^2, \end{aligned}$$

and $S(\cdot)$ denote the sigmoid function of neural firing,

$$\mathrm{S}(\nu) = \frac{\nu_{\max}}{1 + \exp(r(\nu_0 - \nu))}, \tag{1}$$

where a^{-1} and b^{-1} are known time constants, r is a known scale constant and $\nu_{\max}$ is a known maximum firing rate. Let $z(t)$ denote external inputs. Then, the Jansen-Rit model can be described by differential equations below:

$$\begin{aligned} L_a x_0(t) &= AaS(x_1(t) - x_2(t)), \\ L_a x_1(t) &= Aa(Kz(t) + 0.8CS(Cx_0(t))). \\ L_b x_2(t) &= 0.25BbCS(0.25Cx_0(t)) \end{aligned}$$

with output $y(t) = x_1(t) - x_2(t)$. Here, the parameters A and B stand for the average excitatory and inhibitory synaptic gains respectively, the parameter C governs the strengths of the synaptic connectivity between the three neural populations, $C_1 = C, C_2 = 0.8C$, $C_3 = 0.25C$ and $C_4 = 0.25C$ (Cabrera-Álvarez et al., 2023).

Although the Jansen-Rit equations are derived from biological principles, it is not completely determined by these principles. So, the model contains unknown quantities (A, B, C, K, ν_0) that must be inferred from partially observed data before it can be used for prediction or decision making. Estimating the Jansen-Rit model is highly dependent on its structural identifiability. Non-identifiable parameters can create a flat likelihood landscape, leading to biased, inaccurate estimates of the parameters. Moreover, estimates of identifiable parameters can be biased if other parameters are non-identifiable (Heinrich et al., 2025). Here, a parametric system is said to be structurally identifiable if any change in the parameter results in a change in the output. Similarly, we define structural observability for hidden states.

We identify non-identifiable parameters in the model via symmetries of differential equations. Note that under the following scaling transformations

$$\begin{aligned} A &\longrightarrow \lambda A, B \longrightarrow \lambda B, C \longrightarrow \lambda^{-1} C, K \longrightarrow \lambda^{-1} K, \nu_0 \longrightarrow \nu_0, \\ x_0 &\longrightarrow \lambda x_0, x_1 \longrightarrow x_1, x_2 \longrightarrow x_2, \end{aligned}$$

the model preserves its output. The non-identifiability arises because the unknown x_0, B, C, K can absorb arbitrary changes in A, making simultaneous recovery of these

parameters impossible. The symmetry also suggests a reparametrisation scheme for the Jansen-Rit model: $B_A = A/B$, $C_A = AC$, $K_A = AK$, $\nu_0 = \nu_0$, $x_{A0} = x_0/A$, $x_1 = x_1$ and $x_2 = x_2$. After the reparametrisation, we have the form:

$$\begin{aligned} L_a x_{A0}(t) &= aS(x_1(t) - x_2(t)), \\ L_a x_1(t) &= a[K_A z(t) + 0.8 C_A S(C_A x_{A0}(t))], \\ L_b x_2(t) &= 0.25 B_A^{-1} b C_A S(0.25 C_A x_{A0}(t)) \end{aligned} \tag{2}$$

with the same output $y = x_1 - x_2$ as before. The model (2) is called canonical form of the Jansen-Rit model. In this form, fixing $A = 1$ for the synaptic gain of the excitatory neuron population, B_A^{-1} is the synaptic gain of inhibitory neuron populations, C_A measures the strengths of local connectivity between the populations produced by pyramidals, and K_A measures the synaptic effect of the external inputs on the excitatory neuron population.

Following Yates et al. (2009), parameter values (B_A, C_A, K_A, ν_0) and $(B_A^*, C_A^*, K_A^*, \nu_0^*)$ are called equivalent, written $(B_A, C_A, K_A, \nu_0) \sim (B_A^*, C_A^*, K_A^*, \nu_0^*)$ if there exists a positive constant τ such that the associated outputs y and y^* have the same values over time interval $[0, \tau]$. We call a Jansen-Rit model locally structurally identifiable if there exists a small parametric ball in which any two different parameter values are non-equivalent. In the Appendix A, by using the symbolic computing software SIAN (Hong et al., 2019), we show that the canonical Jansen-Rit model is locally structurally identifiable under Condition (**C0**). By a Lie symmetry analysis in S1, the Online Supplementary Material, we also show that the canonical Jansen-Rit model has no non-trival Lie symmetry. In particular, we reveal that there is no parameter A anywhere in the reparameterised system. This is consistent with the recent experimental results in Hauke et al. (2025). In practice, fixing A at a typical value has no effects on estimating (B_A, C_A, K_A, ν_0).

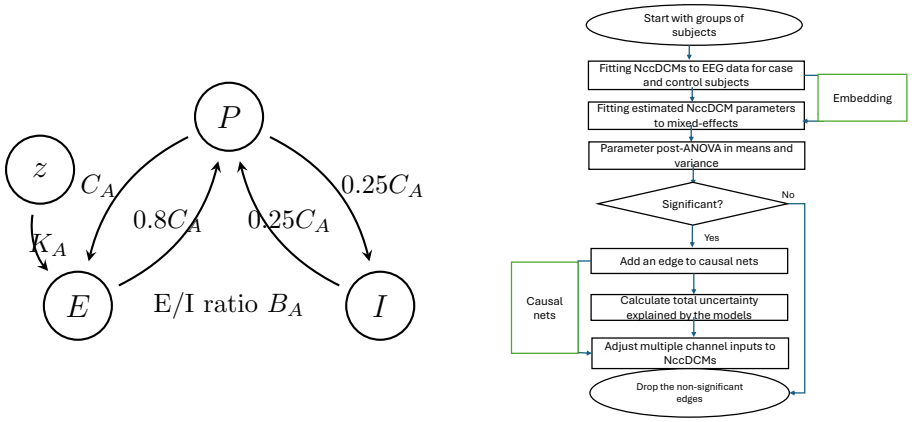


Figure 2. Overview of the NccDCM analysis. Left: Jansen-Rit node model, showing the interactions between local neuron populations Pyramidal (P), Excitatory (E), and Inhibitory (I) under external input (z). Right: The flowchart of the NccDCM analysis.

To model grouped EEG data, more notations are needed. Let $k = 0, 1$ denote group indices and N_k the size of group k. We collected J time series over the time interval $[0, T]$ from each subject using an EEG system. These time series are either sensor time series or source time series obtained from source localisation. For each subject, the time series are partitioned into Q segments, corresponding to disjoint time

intervals. In the epilepsy study, these intervals correspond to seizure or pre-seizure periods as shaded in the time series plots of Figure 1. In the following, these time series are treated as the emissions of a J-node network, each node emits a time series.

Let $x_{A0nq}^{(i)k}, x_{1nq}^{(i)k}$, and $x_{2nq}^{(i)k}$ denote the hidden states of the pyramidal, excitatory interneuron, and inhibitory interneuron populations, respectively, associated with the observed time series $y_{nq}^{(i)k}(t)$ at node i, for subject n in group k during interval q. For notational simplicity, the indices k, n, and q are suppressed when unambiguous. The hidden states evolve according to three coupled second-order ordinary differential equations describing the mean membrane-potential dynamics driven by incoming firing rates within the neural mass. These rates are transformed by the sigmoid activation function in equation (1). The second-order system can equivalently be written as a coupled system of first-order differential equations. After suppressing the indices (k, n, q) from the model parameters, the dynamics of the three neuronal populations at node i, subject to inputs from other nodes and background noise, can be expressed as

$$\begin{aligned}
dx_{A0}^{(i)}(t) &= x_{A3}^{(i)}(t)\,dt, \quad dx_1^{(i)}(t) = x_4^{(i)}(t)\,dt, \quad dx_2^{(i)}(t) = x_5^{(i)}(t)\,dt, \\
dx_{A3}^{(i)}(t) &= \left[a\,\mathrm{S}^{(i|j)}\left(y^{(i)}(t)\right) - 2a\,x_{A3}^{(i)}(t) - a^2\,x_{A0}^{(i)}(t)\right]dt, \\
dx_4^{(i)}(t) &= \Big\{a\left[K_A^{(i|j)}\,\mathrm{S}^{(i|j)}\left(y^{(j)}(t)\right) + 0.8C_A^{(i|j)}\,\mathrm{S}^{(i|j)}\left(C_A^{(i|j)}\,x_{A0}^{(i)}(t)\right)\right] \\
&\quad -2a\,x_4^{(i)}(t) - a^2\,x_1^{(i)}(t)\Big\}\,dt + ad\tilde{\boldsymbol{w}}^{(i|j)}(t), \\
dx_5^{(i)}(t) &= \left[(1/B_A^{(i|j)})b0.25\,C_A^{(i|j)}\,\mathrm{S}^{(i|j)}\left(0.25C_A^{(i|j)}\,x_{A0}^{(i)}(t)\right) - 2b\,x_5^{(i)}(t) - b^2\,x_2^{(i)}(t)\right]dt.
\end{aligned} \tag{3}$$

The output of node i is given by $y^{(i)}(t) = x_1^{(i)}(t) - x_2^{(i)}(t)$. For node i, subject n in group k, and time interval q, after suppressing the indices (k, n, q) from the model parameters, we denote $\sigma_A^{(i|j)} = A^{(i|j)}\sigma^{(i)}$ and define the multiple stochastic input $\tilde{\boldsymbol{w}}^{(i)}(t)$ in the above model by

$$\begin{aligned}
d\tilde{\boldsymbol{w}}^{(i|j)}(t) &= \left[\sum_{j_1 \neq i,j} K_A^{(i|j_1)}\,\mathrm{S}^{(i|j_1)}\left(y^{(j_1)}(t)\right)\right]dt + \sigma_A^{(i|j)}d\boldsymbol{w}^{(i)}(t), \\
\mathrm{S}^{(i|j)}(x(t)) &= \frac{\nu_{\max}}{1 + \exp(r(\nu_0^{(i|j)} - x(t)))},
\end{aligned}$$

where $\boldsymbol{w}^{(i)}(t)$ is a Gaussian noise process, $K_A^{(i|j)}$ is a signal transmission rate from node j to node i, $\sigma_A^{(i|j)}$ is the background noise level, and $\nu_0^{(i|j)}$ is the value for which a 50% of the maximum firing rate is attained by neurons in node i. Let the parameters $B_A^{(i|j)}$ and $C_A^{(i|j)}$ be the values of the parameters B_A, C_A of the neural populations within node i in the presence of input from node j.

Under the assumption that the model parameters are identical across all nodes and all subjects, Forrester et al. (2020) introduced a similar model to describe the modular evolution of the neural populations in each node in the presence of external inputs from the remaining nodes. However, their homogeneity assumption is found not valid later in our EEG data analysis. Our work helps bridge the gap between theory and practice by extending Forrester et al. (2020) to the setting where the model parameters are allowed to vary across nodes and subjects. The equations are solved sequentially for node $i = 1, \ldots, J$, fixing the parameters associated with the remaining nodes. In light of Cabrera-Álvarez et al. (2023), Grimbert and Faugeras (2006), Ditlevsen et al. (2025) and Ableidinger et al. (2017), we set the ranges of these parameters in Table 1.

Table 1. Ranges of biophysical parameters in ccDCM.

Parameter	Description	Value
$B_A^{(i\|j)}$	Ratio of average excitatory-inhibitory synaptic gains	$[0.1, 1]$
a^{-1}	Time constant of excitatory postsynaptic potential	10 ms
b^{-1}	Time constant of inhibitory postsynaptic potential	20 ms
$C_A^{(i\|j)}$	Scaled avg. number of synapses between the populations	$[0, 2500]mV$
$K_A^{(i\|j)}$	Scaled edge from node j to node i	$[0, 2500]$
$v_{\max}$	Maximum firing rate of the neural populations	$5s^{-1}$
$v_0^{(i\|j)}$	Value for which 50% of the maximum firing rate is attained	$[0, 20]$ mV
r	Slope of the act. function at $v_0^{(i\|j)}$	0.56 mV^{-1}
σ_A	Noise level in the region	$[5, 400]$

Let I_k denote the number of active inputs to node k with the output time series $\boldsymbol{y}_k$. Assume that

(**C0**): the data matrix $(\boldsymbol{y}_i)_{i\neq k, i\in I_k}$ has the full column rank.

Then, under the condition **(C0)**, the resulting ccDCMs is shown to be locally weakly identifiable in the Appendix A. A Lie symmetry analysis in S1, the Online Supplementary Material proves that there are no non-trival Lie symmetry on the states $x_{A0}(t), x_1(t), x_2(t)$.

To summarise, in the ccDCM of each node, the dynamic behavior of three neural populations are determined by the above stochastic differential equations. These populations are interconnected through the coupling parameters in the equations, which determine the average evolution rates of the postsynaptic potential. The extrinsic parameter $K^{(i|j)}$ regulates the influence of neural populations in node j on the excitation of neural populations in node i and thus determine the edge between nodes i and j. A large $K^{(i|j)}$ increases the strength of neural synchronization, changing oscillatory activity or pathological dynamics such as epileptic discharges, while decreasing $K^{(i|j)}$ reduces inter-regional coupling, giving rise to more localised dynamics. The above model supports oscillations that relate to input neural rhythms, such as the well-known, alpha, beta, and gamma brain rhythms, and also irregular, epileptic-like activity. It allows us to investigate how neural populations interact under both deterministic and stochastic settings, and how coupling between nodes modulates large-scale brain rhythms observed in E/MEG recordings (Forrester et al., 2020). As the nonlinear sigmoid function $\mathrm{S}^{(i|j)}$ is globally Lipschitz continuous, it follows from Theorem 6.2.2 of Arnold (1974) that the Eq. (3) has a path-wise unique and $\mathcal{F}_t$-adapted solution. In the real data analysis, we fix a, b, $v_{\max}$ and r at their typical values. It is a common practice to rescale the standardised time series data by a constant before fitting a NccDCM (see the software MNE 1.12). This can be further refined by letting r be estimated from the data. Furthermore, it follows from the Appendix A that under the condition (**C0**), the ccDCMs are locally structurally identifiable.

Mixed-effects models for deriving differential DCMs. To facilitate the description of the nested parameters in different groups, we write them in vectors:

$$\boldsymbol{\theta}_{nq}^{(i|j)k} = (\theta_{nq1}^{(i|j)k}, ..., \theta_{nq4}^{(i|j)k}, \theta_{nq5}^{(i)k})^T = (B_{Anq}^{(i|j)k}, C_{Anq}^{(i|j)k}, K_{Anq}^{(i|j)k}, \nu_{0nq}^{(i|j)k}, \sigma_{Anq}^{(i)k})^T.$$

We assume that these scaled parameters follow the mixed-effects model

$$\boldsymbol{\theta}_{nqc}^{(i|j)k} = \boldsymbol{\delta}_{qc} + \boldsymbol{\alpha}_{qc}^{(k)} + \boldsymbol{\beta}_{qc}^{(i|j)} + c_{01}\boldsymbol{\alpha}_{qc}^{(k)} \odot \boldsymbol{\beta}_{qc}^{(i|j)} + \boldsymbol{r}_{nqc}^{(i|j)k}, \quad k = 0, 1; \quad q = 1, ..., Q, \qquad (4)$$

where $\odot$ denotes the component-wise multiplication, $\boldsymbol{\alpha}_{qc}^{(1)} + \boldsymbol{\alpha}_{qc}^{(2)} = 0$, $\sum_{i\neq j} \boldsymbol{\beta}_{qc}^{(i|j)} = 0$, and c_{01} is the coefficient satisfying $c_{01}\sum_q |\beta_{qc}^{(i|j)} + 1| \neq 0$ and $c_{01}\sum_q |\alpha_{qc}^{(i|j)} + 1| \neq 0$.

2.2 Algorithms

Simultaneously estimating parameters for all nodes in the model (3) is difficult as the model is ill-posed. To tackle the issue, we first decompose the large-scale dynamic causal model into J blocks of ccDCMs, where edges in each ccDCM are estimated by treating one node as an input to another. However, brain networks are inherently multivariate, where node outputs are strongly dependent. In particular, EEG sensor data may be spatially degenerate, such that the mapping from bipolar data to reference monopolar data is not one-to-one. As a result, a strong extrinsic coupling between nodes via parameter K_A can be confounded with local parameters (B_A, C_A, ν_0). A screening step to identify marginally significant edges in differential DCM nets at a pre-selected level can help reduce the confounding. The resulting networks are further adjusted for multiple input effects. The details are as follows.

Estimating ccDCMs. We fit the neural mass differential equations in Eq. (3) to the time series data for each pair of nodes in three steps below.

Step 1: Solve the neural mass differential equations for node pair (i, j). For a pairwise screen, we use a misspecified model by ignoring the bias term in $d\tilde{\boldsymbol{w}}_{nq}^{(i|j)k}(t)$. We compute the solutions of the neural mass differential equations iteratively over a sequence of time-instances with the help of the Chen-Fliess expansion in terms of iterative Stratonovich integrals (Fliess, 1981). In Section 3.4.2 and S13, the Online Supplementary Materials, we show the advantage of the Chen-Fliess expansion method over the commonly used Euler-Maruyama (EM) and the stochastic RK in terms of ARE.

For subject n in the group k, the time series emitted by node i during the q-th period are partitioned into D blocks, each containing b_q time-instants $t_{0:b_q-1}^d = (t_0^d, ..., t_{b_q-1}^d)$. The time series for the d-th block is given by $\boldsymbol{Y}_{nq}^{(i)k}(t_{0:b_q-1}^d) = \left(y_{nq}^{(i)k}(t_0^d), y_{nq}^{(i)}(t_1^d), ..., y_{nq}^{(i)}(t_{b_q-1}^d)\right)$, $d = 1, 2, ..., D$. Let $x_{snq}^{(i)k}(t), s = 0, 1, 2$ denote the corresponding three hidden neural states. Let $-\{A3\} = \{A0, 1, 2, 4, 5\}$, $-\{1, 4\} = \{A0, 2, 3, 5\}$ and $x_{Inq}^{(i)k}(t) = (x_{snq}^{(i)k}(t))_{s\in I}$. Denote the truncated Chen-Fliess expansions of the conditional expectations of $y_{nq}^{(i)k}(t)$ and $x_{snq}^{(i)k}(t)$, $s = A0, 2, 3, 5$, given by the forms

$$
\begin{aligned}
y_{nq}^{(i)k}(t_m^d) &\approx g\left(t_m^d | x_{-\{A3\}nq}^{(i)k}(t_0^d), y_{nq}^{(i)k}(t_0^d), \dot{y}_{nq}^{(i)k}(t_0^d), y_{nq}^{(j)k}(t_0^d), \boldsymbol{\theta}_{nq}^{(i|j)k}\right) \\
x_{A0nq}^{(i)k}(t_0^d) &\approx g_0\left(t_{b_q-1}^{d-1} | x_{\{A0,A3\}nq}^{(i)k}(t_0^{d-1}), y_{nq}^{(i)k}(t_0^{d-1}), \dot{y}_{nq}^{(i)k}(t_0^{d-1}), \boldsymbol{\theta}_{nq}^{(i|j)k}\right), \\
x_{2nq}^{(i)k}(t_0^d) &\approx g_2\left(t_{b_q-1}^{d-1} | x_{-\{1,4\}nq}^{(i)k}(t_0^{d-1}), \boldsymbol{\theta}_{nq}^{(i|j)k}\right), \\
x_{A3nq}^{(i)k}(t_0^d) &\approx g_3\left(t_{b_q-1}^{d-1} | x_{\{A0,A3\}nq}^{(i)k}(t_0^{d-1}), y_{nq}^{(i)k}(t_0^{d-1}), \dot{y}_{nq}^{(i)k}(t_0^{d-1}), \boldsymbol{\theta}_{nq}^{(i|j)k}\right), \\
x_{5nq}^{(i)k}(t_0^d) &\approx g_5\left(t_{b_q-1}^{d-1} | x_{-\{1,4\}nq}^{(i)k}(t_0^{d-1}), \boldsymbol{\theta}_{nq}^{(i|j)k}\right),
\end{aligned} \tag{5}
$$

where $g(t|\cdot), g_0(t|\cdot), g_2(t|\cdot), g_3(t|\cdot)$ and $g_5(t|\cdot)$ are defined in the Appendix B. Then the truncated Chen-Fliess expansions of the above Jansen-Rit differential system can be written as a conditionally deterministic function, say g or g_m, plus a stochastic error term. Moreover, in the expansions of $x_{A0nq}^{(i)k}, x_{2nq}^{(i)k}(t)$, the stochastic error terms, only appearing in the terms of orders larger than 3, are bounded by $|o_p((t-t_0)^4)|$, while in the expansion of $y_{nq}^{(i)k}(t)$, the stochastic terms, not appearing in the term of order 1, is of the form $\sigma_{Anq}^{(i|j)k} a \int_{t_0}^t (t-s) d\tilde{\boldsymbol{w}}_{nq}^{(i)k}(s) + o_p((t-t_0)^3)$. Therefore, these approximate

functions of the conditional expectations in (5) provide the best one-step predictions of $y_{nq}^{(i)k}(t)$ and $x_{snq}^{(i)k}(t)$.

Step 2: Construct loss functions. Let $t_0^d = t_{b_q-1}^{d-1}$. We randomly initiate $x_{-\{1,4\}nq}^{(i)k}(t_0^1)$ by the standard multivariate norm with mean value $(0,...,0)$. Given the observed D block data, we can iteratively make one-step forward predictions of both the hidden states and the node outputs over time-grids by using the truncated Chen-Fliess expansions and the difference operator as follows. For the first block,

$$\begin{aligned}
\dot{y}_{nq}^{(i)k}(t_0^1) &\approx (y_{nq}^{(i)k}(t_1^1) - y_{nq}^{(i)k}(t_0^1))/(t_1^1 - t_0^1), \\
\widehat{y}_{nq}^{(i)k}(t_m^1) &\approx g\left(t_m^1 | x_{-\{A3\}nq}^{(i)k}(t_0^1), y_{nq}^{(i)k}(t_0^1), \dot{y}_{nq}^{(i)k}(t_0^1), y_{nq}^{(j)k}(t_0^1), \boldsymbol{\theta}_{nq}^{(i|j)k}\right),
\end{aligned}$$

where $x_{\{A0:5\}nq}^{(i)k}(t_0^1) = (0, y_{nq}^{(i)k}(t_0^1), 0, 0, \dot{y}_{nq}^{(i)k}(t_0^1), 0)^T$. In general, for $m = 1, ..., b_q - 1$ and $d = 2, ..., D$,

$$\begin{aligned}
\dot{y}_{nq}^{(i)k}(t_0^d) &\approx (y_{nq}^{(i)k}(t_1^d) - y_{nq}^{(i)k}(t_0^d))/(t_1^d - t_0^d), \\
\widehat{y}_{nq}^{(i)k}(t_m^d) &\approx g\left(t_m^d | x_{-\{A3\}nq}^{(i)k}(t_0^d), y_{nq}^{(i)k}(t_0^d), \dot{y}_{nq}^{(i)k}(t_0^d), y_{nq}^{(j)k}(t_0^d), \boldsymbol{\theta}_{nq}^{(i|j)k}\right), \\
\widehat{x}_{A0nq}^{(i)k}(t_0^d) &\approx g_0\left(t_{b_q-1}^{d-1} | x_{\{A0,A3\}nq}^{(i)k}(t_0^{d-1}), y_{nq}^{(i)k}(t_0^{d-1}), \dot{y}_{nq}^{(i)}(t_0^{d-1}), \boldsymbol{\theta}_{nq}^{(i|j)k}\right), \\
\widehat{x}_{2nq}^{(i)k}(t_0^d) &\approx g_2\left(t_{b_q-1}^{d-1} | x_{-\{1,4\}nq}^{(i)k}(t_0^{d-1}), \boldsymbol{\theta}_{nq}^{(i|j)k}\right), \\
\widehat{x}_{1nq}^{(i)k}(t_0^d) &= y_{nq}^{(i)k}(t_0^d) + \widehat{x}_{2nq}^{(i)k}(t_0^d), \\
\widehat{x}_{A3nq}^{(i)k}(t_0^d) &\approx g_3\left(t_{b_q-1}^{d-1} | x_{\{A0,A3\}nq}^{(i)k}(t_0^{d-1}), y_{nq}^{(i)k}(t_0^{d-1}), \dot{y}_{nq}^{(i)}(t_0^{d-1}), \boldsymbol{\theta}_{nq}^{(i|j)k}\right), \\
\widehat{x}_{5nq}^{(i)k}(t_0^d) &\approx g_5\left(t_{b_q-1}^{d-1} | x_{-\{1,4\}nq}^{(i)k}(t_0^{d-1}), \boldsymbol{\theta}_{nq}^{(i|j)k}\right), \\
\widehat{x}_{4nq}^{(i)k}(t_0^d) &= \dot{y}_{nq}^{(i)k}(t_0^d) + \widehat{x}_{5nq}^{(i)k}(t_0^d).
\end{aligned}$$

For the subject n in group k and node pair (i, j) the loss function/negative log-likelihood function in node i, with a regressive input from node j, can be written as

$$L_{nq}^{(i|j)k}(\boldsymbol{\theta}_{nq}^{(i|j)k}) = \frac{D}{2}\log 2\pi + \frac{1}{2}\sum_{d=1}^{D}\log G_{nqd}^{(i|j)k} + \frac{1}{2}\sum_{d=1}^{D}\frac{\left(y_{nq}^{(i)k}(t_{b_q-1}^d) - \widehat{y}_{nq}^{(i)k}(t_{b_q-1}^d)\right)^2}{G_{nqd}^{(i|j)k}},$$

where $G_{nqd}^{(i|j)k} = (\sigma_{Anq}^{(i)k}a)^2(t_{b_q-1}^d - t_0^d)^3/6$.

Step 3: Minimise the loss function by genetic evolutionary algorithms. Minimising the above loss function with respect to the model parameters, we obtain the estimate $\widehat{\boldsymbol{\theta}}_{nq}^{(i|j)k}$. Note that S2, the Online Supplementary Material demonstrates that the above non-convex loss function contains multiple local minima as only partial observations on hidden states are available in dynamic causal modelling. We employ a genetic evolutionary algorithm of Zhang and Sanderson (2009), called adaptive differential evolution with optional external archive (JADE), to search for global minima. JADE evolves a population of candidate solutions through mutation, crossover, and selection, aiming to explore the landscape of the loss function around a global minimum. See S2, the Online Supplementary Material for the further details of the JADE.

To mitigate the systematic bias present in the above estimation procedure, we further employ a bootstrap bias correction method. See S4, the Online Supplementary Material.

Identify differential causal nets for case and control groups. Having obtained estimated networks of ccDCMs for cases and controls respectively, we want to find differential causal nets by contrasting cases and controls in terms of the estimated within-node parameters $(\widehat{B}_{Anq}^{(i|j)k}, \widehat{C}_{Anq}^{(i|j)k}, \widehat{\nu}_{0nq}^{(i|j)k})$ and the estimated between-node parameters $\widehat{K}_{Anq}^{(i|j)k}$ respectively. The procedure is implemented into two steps: testing and screening followed by the adjustment for multiple-input effects. In the differential causal net for the c-th component of the estimated parameter vector, we assign a directed link from node j to node i if the estimated values $(\widehat{\boldsymbol{\theta}}_{nqc}^{(i|j)1})_{1\le q\le Q}, 1 \le n \le N_1$, in the cases are significantly different from their counterparts in the controls.

Step 1: Hypothesis testing. In order to test for the significant differences between cases and controls, we treat the case-control memberships and node memberships as factors in the following two-factor mixed-effects model

$$\widehat{\boldsymbol{\theta}}_{nq}^{(i|j)k} = \boldsymbol{\delta}_q + \boldsymbol{\alpha}_q^{(k)} + \boldsymbol{\beta}_q^{(i|j)} + c_{01}\boldsymbol{\alpha}_q^{(k)} \odot \boldsymbol{\beta}_q^{(i|j)} + \boldsymbol{r}_{nq}^{(i|j)k}, \quad k = 0, 1; \quad q = 1, ..., Q, \tag{6}$$

where $\odot$ denotes the component-wise multiplication and we write each estimated parameter as the summation of a baseline, fixed-factors related to case-control memberships and nodes, and subject-specific random factors. Here, the significance of the case-control factor or the interaction factor between the case-control and nodes implies that case-control memberships make a contribution to the group differences while the significance of the node-factor shows that the group difference should be accounted in study. The significance of random-effects (e.g., the difference of the variations across groups) implies that subject-specific variations play a role in differencing the groups from each other.

To make the above model identifiable, we assume that $\boldsymbol{\alpha}_q^{(1)} + \boldsymbol{\alpha}_q^{(2)} = 0$, $\sum_{i\neq j} \boldsymbol{\beta}_q^{(i|j)} = 0$, and c_{01} is the coefficient satisfying $c_{01}\sum_q |\beta_{qc}^{(i|j)} + 1| \neq 0$ and $c_{01}\sum_q |\alpha_{qc}^{(i|j)} + 1| \neq 0$. Under the assumption that the parameters for different subjects are independent of each other, the expected Manhattan distances between the groups and within the groups admit

$$\begin{aligned} E\sum_q |\theta_{nqc}^{(i|j)1} - \theta_{mqc}^{(i|j)0}| &= \sum_q E|r_{1qc}^{(i|j)0} - r_{1qc}^{(i|j)1}|, \text{ under } H_0 \\ E\sum_q |\theta_{nqc}^{(i|j)k} - \theta_{mqc}^{(i|j)k}| &= \sum_q E|\theta_{nqc}^{(i|j)k} - \theta_{mqc}^{(i|j)k}|, \quad k = 0, 1. \end{aligned} \tag{7}$$

Approximating the expectations in equation (7) by the U-statistics Lee (1990), we have the following statistics: For $1 \le c \le 5$, $k = 0, 1$, and $1 \le i \neq j \le J$,

$$\begin{aligned} \sigma_{ckk}^{(i|j)} &= \frac{2}{N_k(N_k-1)Q} \sum_{1\le m<n\le N_k} \sum_{1\le q\le Q} |\theta_{mqc}^{(i|j)k} - \theta_{nqc}^{(i|j)k}|, \\ \sigma_{c01}^{(i|j)} &= \frac{1}{N_0 N_1 Q} \sum_{1\le m\le N_0, 1\le n\le N_1} \sum_{1\le q\le Q} |\theta_{mqc}^{(i|j)0} - \theta_{nqc}^{(i|j)1}| \end{aligned}$$

which are the empirical versions of the within-group and between group average distances. For $1 \le c \le 5$ (estimable parameters), to test the above null hypothesis H_0, we consider the test statistics $\boldsymbol{F}^{(i|j)} = (F_1^{(i|j)}, ..., F_5^{(i|j)})^T$ with

$$F_c^{(i|j)} = \frac{(N_0 + N_1)\sigma_{c01}^{(i|j)}}{N_0\sigma_{c00}^{(i|j)} + N_1\sigma_{c11}^{(i|j)}}.$$

We permuted the case-control memberships M times with default value $M = 5999$. For each permutation, we recomputed the value of $F_c^{(i|j)}$, obtaining M permuted F-values. The permuted p-values p_c is then defined by counting the proportion of the permuted F-values that are less than the observed value of $F_c^{(i|j)}$. We also run a permuted F-test for differences in the variances of the random factors of patients and controls. Similarly, we test the null hypothesis that $\beta_{qc}^{(i|j)} = 0$ for all (i, j).

Pairwise screening. We select node pairs that have p-values at the level 0.01 or at a more stringent level 0.005 such that for each node, the multiple inputs meet the identifiability condition (**C0**).

Step 2: Adjustment for multiple input effects. In the above pairwise screening, we only account for the marginal effects of multiple input nodes on a target node. This may result in confounding between direct and indirect effects. To mitigate the problem, we adjust the multiple-input effects for the node pairs that survived in the previous screening step below. For each target node, we identify input nodes that are linked to it and survived in the previous screening step. Taking these nodes as multiple inputs, we refit a ccDCM to the time series emitted from the target node for each subject in the two groups. We then perform the mixed-effects model-based tests for differences in the estimated model parameters between the groups. We replace the pairwise p-values by these new p-values. Finally, the Benjamini–Hochberg (BH) correction is applied to the resulting p-values across all tested node pairs to control the false discovery rate. We retain the node edges with BH-adjusted p-values less than or equal to 0.05.

Identifying differential causal nets for subjects paired groups. Understanding differences of epileptic patients between the preictal period and ictal period is crucial for gaining deeper insights into the mechanisms underlying epileptic seizures. While most studies focus on the ictal states, examining the transition from the preictal phase to the ictal phase can reveal early neural signatures that precede seizure onset. Identifying such signatures are important not only for elucidating pathophysiological dynamics of epilepsy but also for improving the characterisation of the transition from the preictal to the ictal and thus developing early intervention strategies. We performed multiple permutation-based Wilcoxon signed-rank tests for significant NccDCM changes from the preictal period to the ictal period. See S3, the Online Supplementary Material for the details.

2.3 Epilepsy dataset

Data description. The dataset contains EEG recordings in 18 EEG channels for 21 subjects who underwent EEG tests for epilepsy. Longitudinal bipolar montage is used for channel configuration for the EEG recording, and the considered channels are FP2F4, F4C4, C4P4, P4O2, FP1F3, F3C3, C3P3, P3O1, FP2F8, F8T4, T4T6, T6O2, FP1F7, F7T3, T3T5, T5O1, FZCZ, CZPZ, and BP3REF (18 EEG channels and 1 ECG channel). See Table S1 in S6, the Online Supplementary Material for the brain regions which the electrode-pairs cover. EEG signals were band-pass filtered between 1–40 Hz and re-referenced implicitly through the bipolar montage. No additional re-referencing was applied to preserve the original spatial derivations. Segmentation information provided by the owner in an accompanying metadata identified ictal intervals via start and end times; all remaining intervals were treated as interictal periods.

We project these channels onto two concentric rings- outer and inner- based on

their spatial positions on the scalp. A sensor-level differential causal net per parameter is built by assigning significant edges to these channels. To test for significant edges, sensor interictal epileptiform discharge (IED) time series from 11 subjects were identified and confirmed by neurologists while the sensor time series from the remaining 10 subjects were free from IEDs. Furthermore, for each of 8 patients, we were able to define 6 segments of IED time series, each of length 20 seconds. These segments were respectively matched by the segments of sensor time series of 10 controls at the same time-periods. The dataset is preprocessed using independent component analysis with six chosen components, embedded in the software dSPM. We divide each sensor time series into epochs, each of size 2 seconds. Here, we aim to analyse this dataset to identify abnormal dynamic causal nets associated with the IED in the sensor and source levels.

Source localisation with bipolar EEG sensor data. EEG sensors record linear mixtures of signals originating from multiple sources in the brain. Although the forward model from source signals to sensor measurements is linear, the nonlinear dynamics of the Jansen–Rit model do not generally commute with this mixing operation. See the Appendix B for the details. Consequently, Jansen–Rit parameters estimated from sensor-level mixtures cannot, in general, be inversed to obtain the corresponding source-level parameters through a linear transformation. So, to construct differential causal nets in the source level, we need to localise the intracranial sources that generated observed EEG measurements on the scalp. This is an ill-posed inverse problem because a very large number of different sources can produce the same potential field on the scalp. For this purpose, we first convert the bipolar sensor measurements to a dataset of the monopolar format, to which the distributed source localisation package dSPM (Michel and Brunet, 2019) is applied. Specifically, let $B_2(t)$ and $B_1(t)$ denote the bipolar and monopolar sensor measurements, respectively, and let M denote the transformation matrix from the monopolar to the bipolar montage: $B_2(t) = MB_1(t)$. Let L denote the lead-field matrix derived from the average surface model in the software FreeSurfer. The monopolar measurements can then be represented by the forward model $B_1(t) = L\beta(t) + E(t)$, where $\beta(t)$ denotes the underlying source time series and $E(t)$ represents error terms. We obtain the following monopolar representation of the measurements from the bipolar data using the Moore–Penrose pseudoinverse of M

$$B_2(t) = ML\beta(t) + ME(t).$$

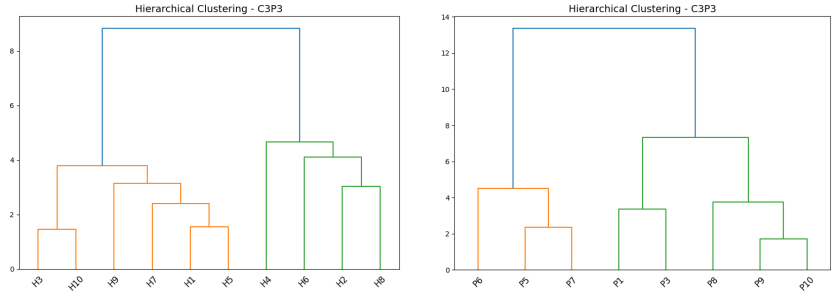
In practice, applying dSPM to $M^+B_2(t)$, we obtain the estimate of source time series, $\widehat{\beta}$, at grid points in the source-space. See S9, the Online Supplementary Material for the theory. We then group the grid points into the 68 cortical regions defined by the Desikan–Killiany (D-K) atlas and using sign-flipped means to average the estimated source time series over the grid points within each region. This results in 68 estimated regional source time series, one for each D-K cortical region. We project these regions/nodes onto three concentric rings— outer, middle, and inner—based on their spatial locations on the cortical surface. See the abbreviations of these region names in S10, the Online Supplementary Material. A source-level differential causal net per parameter is constructed by assigning significant edges to these nodes. Note that the K_A differential causal nets measure the significant difference between the input-output relationships in the case-control groups while the B_A, C_A and ν_0 differential causal nets show impacts of these input-output relationships on the local node differences between the case-control groups.

3 Numerical results

In this section, we aimed to investigate the behavior of the proposed models and to evaluate the performance of the proposed estimating method using simulated data as well as the real epileptic dataset of Fasil et al. (2021). We first conducted sensor- and source-level analyses of the case-control data, followed by analyses of the preictal–ictal data and, finally, simulation studies.

3.1 Sensor differential dynamic causal nets

We began with a subject-heterogeneity analysis for control-case sensor time series data using hierarchical clustering. We took the C3P3 channel as an example. The subject's EEG time series in the C3P3 and other channels were extracted from the same time period across all subjects. We calculated the feature vector of marginal mean, variance and the autocorrelations at the first 50 lags for each subject's EEG time series. Subject-heterogeneity in the C3P3 and other channels was assessed through hierarchical clustering of these feature vectors, where the Euclidean distance based average linkage method was adopted to generate dendrograms. The results for the C3P3 and other channels, displayed in Figure 3 and Figure S3 in Section S8, the Online Supplementary Material, showed evidence of subject-heterogeneity in the case and control groups. These findings gave us a rationale to fit the NccDCM individually for each subject to capture subject-specific dynamics. This approach allowed us to make group-level inferences while accounting for individual variability through a mixed-effects model.



Controls Cases

Figure 3. Dendrograms for clustering subjects based on the time series data derived from the channel C3P3 for the control and case groups.

In the following analysis, we focused on these 8 patients against 10 controls. See a flowchart of the data analysis in Figure 2.

Differential dynamic causal nets in mean and variation. For each pair of channels, we first fit ccDCMs to recorded patients' IED time series and their counterparts in controls respectively, obtaining the estimated values of parameters in the ccDCMs after the corrections of multiple testing and multiple input-effects. We then perform a post-analysis of variance on these estimated parameters to reveal the IED-associated causal patterns in terms of channel excitation-inhibition ratio of synaptic gain, the strength of within-channel connectivity, the strength of between-channel connectivity and the contrasts of synaptic transmission between cases and controls during seizure periods. The manhattan distance was used in these post-analyses of variance. Based on the adjusted $(1-p)$-values, we constructed the adjacency matrix to reflect the degree of difference between the case and control

groups. To facilitate the presentation, we retained only the p-values that are less than 95%. The resulting differential dynamic causal nets are displayed in Figures 4 and 5. In the nets, the directionality was indicated by arrows from an input channel to an output channel. The value of $-\ln(p)$ increased from 3 to 5.5 as colors are ranging from yellow to dark red to yellow. From Figure 4, we revealed the following causal nets:

- B_A: In Figure 4(a)(b), there were 3 and 6 channel pairs to be claimed to differ cases from controls significantly in mean and variance respectively, where the front channels FP1F7, FP1F3, FP2F8 and the back channels C3P3, P3O1 were involved.

- C_A: In Figure 4 (c)(d), there were 2 and 2 channel pairs to be found to have significant differences between cases and controls in mean and variance respectively. Notably, these differences were most pronounced in channel pairs where prefrontal regions served as output channels, including the front channels FP1F3 and FP2F4 and the back channel P3O1.

- K_A: In Figure 4(e)(f), 16 and 10 channel pairs exhibited significant differences between cases and controls in mean and variance respectively. These differences were predominantly observed in the frontal regions, including FP1F7, FP1F3, FP2F8, FZCZ and FP2F4.

- ν_0: In Figure 5(a)(b), there were 11 channel pairs which showed significant differences between cases and controls. The most pronounced differences were in the prefrontal and central-temporal-parietal region, where F7T3, FZCZ, FP1F7, FP1F3, C3P3, T5O1, P4O2 and T4T6 were involved.

- σ_A: In Figure 5(c), there were 4 channel pairs to be significant, where the front channels F4C4, FZCZ and F3C3 were involved.

In summary, for the parameters B_A, C_A, K_A, ν_0 and σ_A, the most pronounced differences between the case and control groups were in channel pairs covering frontal, temporal and back brain regions. In particular, FP1F7, FP1F3, FP2F8, FZCZ and FP2F4 are related to the left and right dorsolateral and ventrolateral prefrontal cortices. This finding highlighted the prominent role of these prefrontal areas in driving network-level alterations. An altered excitation/inhibition (E/I) balance, often involving decreased inhibition, has been identified as a robust biomarker in various neurodegenerative diseases such as Alzheimer's disease (AD) and frontotemporal dementia (FTD). Our analysis suggested a mechanism that epilepsy shared with other neurological diseases that changes in the inhibitory connections in the frontal cortex disrupted normal circuit operations and brain network dynamics. The above findings centred on FP1F7, FP1F3, FP2F8 and FP2F4. Unfortunately, these derivations could also carry most eye-movement and blink artefacts, more so in children. But these artefacts have been removed by independent component analysis in preprocessing EEG recordings. Note also that FP1F3 and F3C3 share F3, whose outputs are coupled by construction with opposite signs. Volume conduction might also add further zero-lag coupling. Therefore, edges between adjacent channels could be partially caused by the montage design. The source level analysis in the next subsection will further support these findings in the prefrontal source space.

Post ANOVA tables for the NccDCM. We treated the fitting of the NccDCM to each subject as a transformation of its EEG measurements into the values of the estimated network parameters, which inherited the uncertainty in the original data. We used the `adonis2` function in the R software to perform post multivariate ANOVA on these values and to test the significance of case-control and channels as two factors.

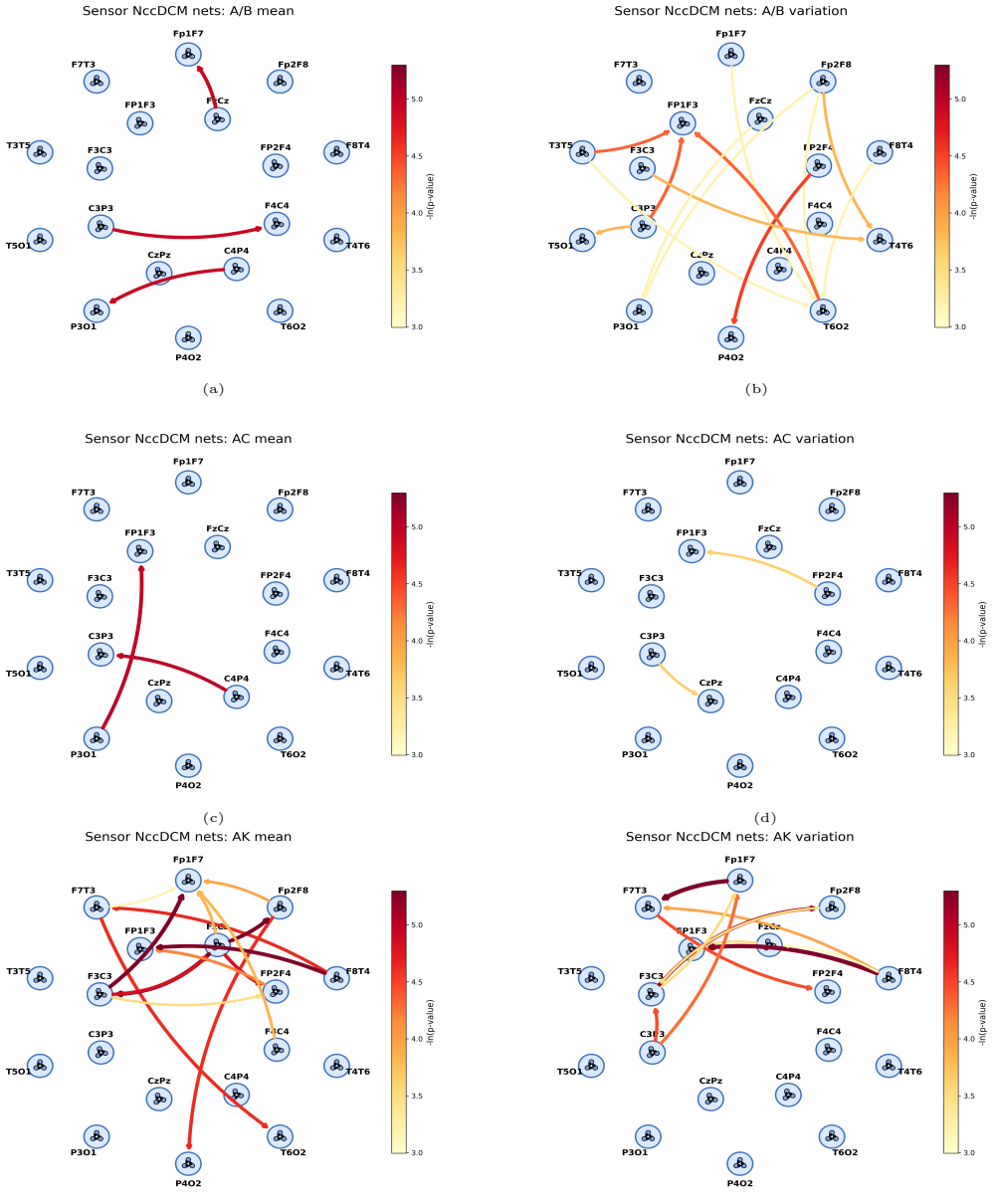


Figure 4. Sensor differential dynamic causal nets in mean and variation for parameters after the adjustments for multiple testing. (a) and (b): $B_A = A/B$. (c) and (d): $C_A = AC$. (e) and (f): $K_A = AK$. The colour bars are in the scale of $-\ln$(p-value), where p-value is BH-adjusted.

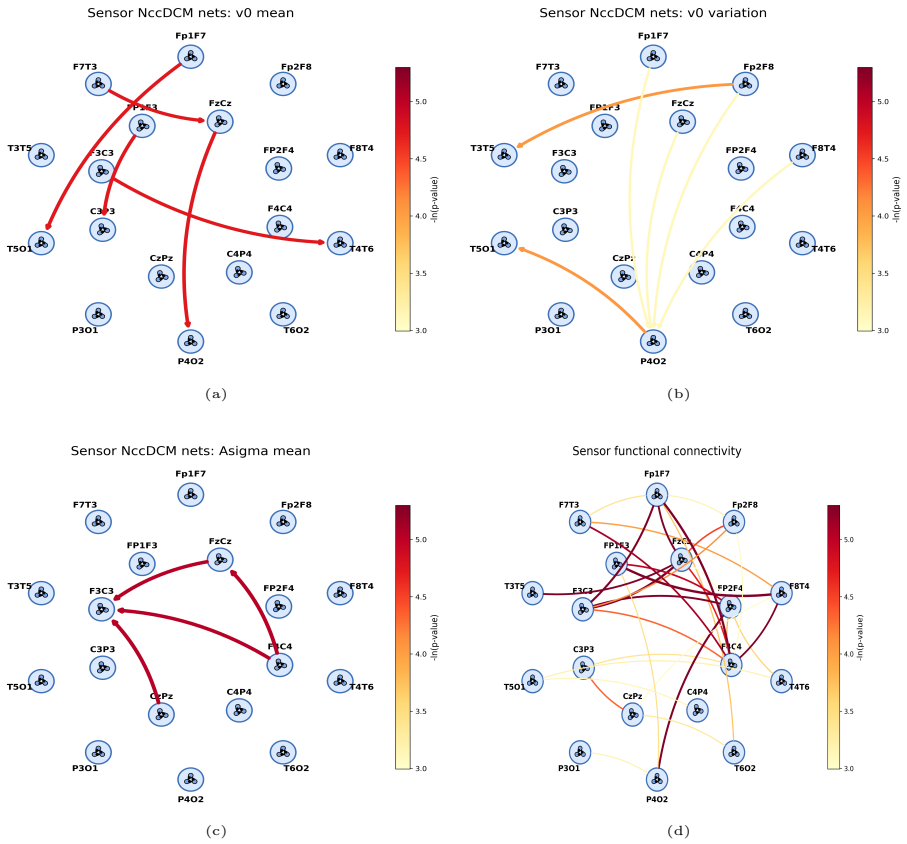


Figure 5. Sensor differential dynamic causal nets in mean and variation for parameters after the adjustments for multiple testing. (a) and (b): ν_0. (c): $\sigma_A = A\sigma$. (d): Sensor differential causal nets using the maximum lagged correlations-based functional connectivities between case and control groups at the level 5%. However, these edges in the nets were not significant at the level of 5% after the adjustments for multiple testing. The colour bars are in the scale of $-\ln$(p-value), where p-value is BH-adjusted.

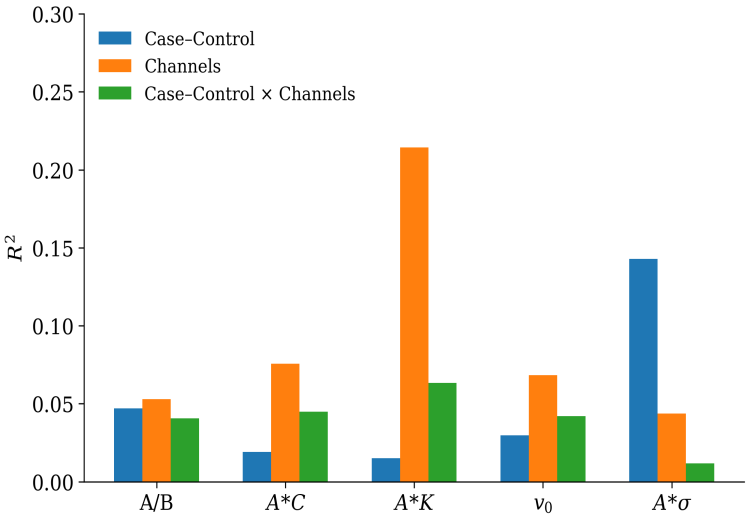


Figure 6. The percentages of uncertainty in the estimated sensor parameters which was explained by the NccDCM-based mixed-effects models, R^2. See Anderson (2017) for the definition of R^2. Here, $B_A = A/B, C_A = AC, K_A = AK$ and $\sigma_A = A\sigma$.

The numerical results are displayed in Figure 6 and in Tables S2 to S6 in the Section S7, the Online Supplementary Material. The results indicated that the identified dynamic causal nets were highly significant for each parameter. The percentages of variations in the estimates explained by the estimated dynamic causal nets, due to Case-Control factor plus its interaction with Channels, for B_A, C_A , K_A, ν_0 and σ_A were respectively 14%, 14%, 29%, 14% and 19.9% in means and 14%, 25.9%, 44%, 17.6% and 0% in variations. The percentages of variations in the estimated parameters, R^2 values confirmed the significant heterogeneity of the network parameters across the channels. For example, around 6.3% (p-value 0.000167) of the variations in the connection rate K_A between regions was due to the channel difference.

Lagged correlation-based functional connectivity analysis. To compare with lagged correlation analysis, we calculated lagged correlation coefficients using the method in S5, the Online Supplementary Material. The analysis revealed 31 channel pairs with p-values less than 0.05 in testing the difference of lagged correlations between the control and case groups as displayed in Figure 5(d). The channel pairs (FP2F4, P4O2), (FP2F4, F3C3), (F4C4, FP1F7), (FP1F3, F8T4), (F3C3, FP1F7), (F3C3, FZCZ), and (FP1F7, FZCZ) with p-values were smaller than 0.005. Among them, channel pairs (F4C4, FP1F7), (F3C3, FP1F7) and (F3C3, FZCZ) were also identified in the K_A based NccDCM analysis. However, unlike the previous NccDCM analysis, there were no significant pairs left at 5% after the BH correction for multiple testing. This showed an advantage of the NccDCM analysis over the conventional functional connectivity analysis. The similar advantage was also found in source level but omitted.

3.2 Source differential dynamic causal nets

We first applied the bipolar source localisation to the Epilepsy EEG dataset to obtain an estimated source time series for each of 68 D-K regions. We fit a NccDCM with 68 nodes to the estimated source time series data. The results, displayed in Figures 8∼9 below.

- The B_A differential causal nets in mean and variation contain a mixture of

temporal, parietal, frontal, occipital regions, on both sides of the brain. The E/I ratio maintains the stability, precise neuronal coding and information processing in neural circuits. The changes of the E/I ratios, due to abnormal input-to-output relationships between these regions, are linked to neurological disruptions generated by epilepsy.

In Yang et al. (2023), significant increases were observed in frequency-specific correlations between the anterior cingulate cortex and the anterior insular cortex, as well as in the excitation/inhibition ratio during seizures. The cortical regions identified in the present causal nets were previously reported as epilepsy-related structural abnormalities in the following brain network systems (Whelan et al., 2018; Larivière et al., 2022): Default Mode Network (DMN) (Posterior cingulate, middle temporal, inferior parietal, entorhinal/medial temporal regions and temporal pole); Frontoparietal/executive control network (Precuneus, superior parietal, inferior parietal, caudal middle frontal, parts of supramarginal and inferior frontal cortex); Dorsal attention network (Superior parietal, inferior parietal and frontal control regions); Sensorimotor network (Precentral and postcentral); Visual network (Pericalcarine, cuneus, lingual, fusiform and lateral occipital); Auditory network (Transverse temporal, banks of superior temporal sulcus, middle temporal, supramarginal and pars orbitalis); Cognitive-control systems (Anterior cingulate, orbitofrontal cortex and frontal regions). The fact that these nets involve regions belonging to the DMN, attention/executive, sensorimotor, visual and auditory systems could indicate alterations in a distributed network-level, rather than simply abnormalities in a single functional system. For example, the insula is anatomically concealed, so as physiologically, with a cryptic role in epilepsy while entorhinal cortex is particularly relevant to temporal-lobe epilepsy. But the other regions can reflect propagation networks, secondary effects, or group-level cortical changes rather than the seizure-onset zone.

- C_A differential causal nets show some fairly coherent epilepsy-network alternation patterns related to the following network systems: Medial temporal (Entorhinal, parahippocampal, temporal pole and fusiform); Temporo-parietal (Superior temporal, supramarginal , superior parietal); Cingulate/salience (Rostral & caudal anterior cingulate, isthmus cingulate and insula); Frontal/executive (Superior frontal, frontal pole, medial/lateral orbitofrontal and pars opercularis/orbitalis); Visual (Lingual, lateral occipital and fusiform).

  Compared to healthy controls, in a few structural MRI analyses, thickness differences of the ipsilateral temporopolar, parahippocampal, entorhinal, and fusiform gyri, contralateral pars triangularis, and bilateral precuneus, superior frontal and caudal middle frontal gyri were observed in left, but not right of the brains with mesial temporal lobe epilepsies and hippocampal sclerosis (Whelan et al., 2018). Note that the parameter C_A scales inter-regional coupling, governing how strongly these regions excite/inhibit one another as demonstrated in Figure 2. Compared to the healthy subjects, the changes of input-to-output relationships between these regions significantly modified regional coupling structures in the brain.

- The K_A differential dynamic causal nets in mean and variation define direct measurements of alterations in a distributed network in terms of input-output coefficients between regions. They were consistent with the findings in a strucure analysis of focal and generalised epilepsy (Larivière et al., 2022). The regions involved were actually reported in a number of epilepsy studies. For example, Ogren et al. (2018) reported abnormalities involving the insula, frontal pole,

lateral orbitofrontal, cingulate, paracentral, postcentral, cuneus and pericalcarine cortices in generalized tonic-clonic seizures. Sammarra et al. (2024) reported cortical-thickness abnormalities in inferior parietal, inferior temporal, middle temporal, temporal pole and isthmus-cingulate regions in mesial temporal lobe epilepsy (MTLE). Wu et al. (2018) also identified several of these regions as altered network features in temporal lobe epilepsy (TLE).

- The ν_0 differential causal nets in mean and variation involve the regions which have also well-documented associations with epilepsy-related structural brain abnormalities, particularly in large ENIGMA-Epilepsy studies (Whelan et al., 2018).

- The σ_A differential causal net in variation implies that regions lh-caudalmiddlefrontal and rh-caudalanteriorcingulate both have a high degree of inputs from other regions. This was consistent with emerging evidences that the left caudal middle frontal gyrus played a role in front networks of the brains with epilepsy (Luo et al., 2014).

Post ANOVA tables for the NccDCM. In the source space, we performed post multivariate ANOVA on the estimated parameter values and to test the significance of case-control and regions as two factors. The results, displayed in Figure 9 and Tables S7–S11, show the region heterogeneity indicated by the significant interactions of two factors.

3.3 Differential dynamic causal nets between preictal and ictal periods

Description of pair-matched data According to the annotations provided by Fasil et al. (2021), in the preictal periods, the epileptic subjects P1, P3 and P5∼P9 were selected with six 20 seconds long preictal segments of time series identified. P2, P4 and P10 were excluded from further analysis as there were no six 20 seconds long preictal segments identified. Note that the preictal-ictal period dataset was period-matched for each subject while the case-control dataset was not pair-matched.

Sensor level analysis. We fitted a NccDCM to the EEG recordings for each subject in the preictal period and the ictal period respectively, followed by the pair-matched post Wilcoxon signed rank test for the period differences. The results illustrated in Figure 10 suggest that, in patients with epilepsy, the NccDCM parameters exhibited noticeable increases from the preictal period to the ictal period. These changes were associated with seizure onset.

- In the σ_A sensor differential dynamic causal net, channels F7T3, C3P3 and FP2F4, with relatively high degree centralities, imply a spatially distributed preictal-to-ictal EEG activity involving bilateral frontotemporal, centro-parietal, and right frontal regions.

- In the C_A sensor differential dynamic causal net, the significant change in the local connection parameter was identified due to the input-output relationship between channels FP2F8 and F3C3.

- In the K_A sensor differential dynamic causal net, the significant changes of input-output coefficients were identified in channel pairs (T6O2,P3O1), (FP1F3, FP2F4) and (CZPZ, F4C4), related to the connections between the frontal and temporal regions.

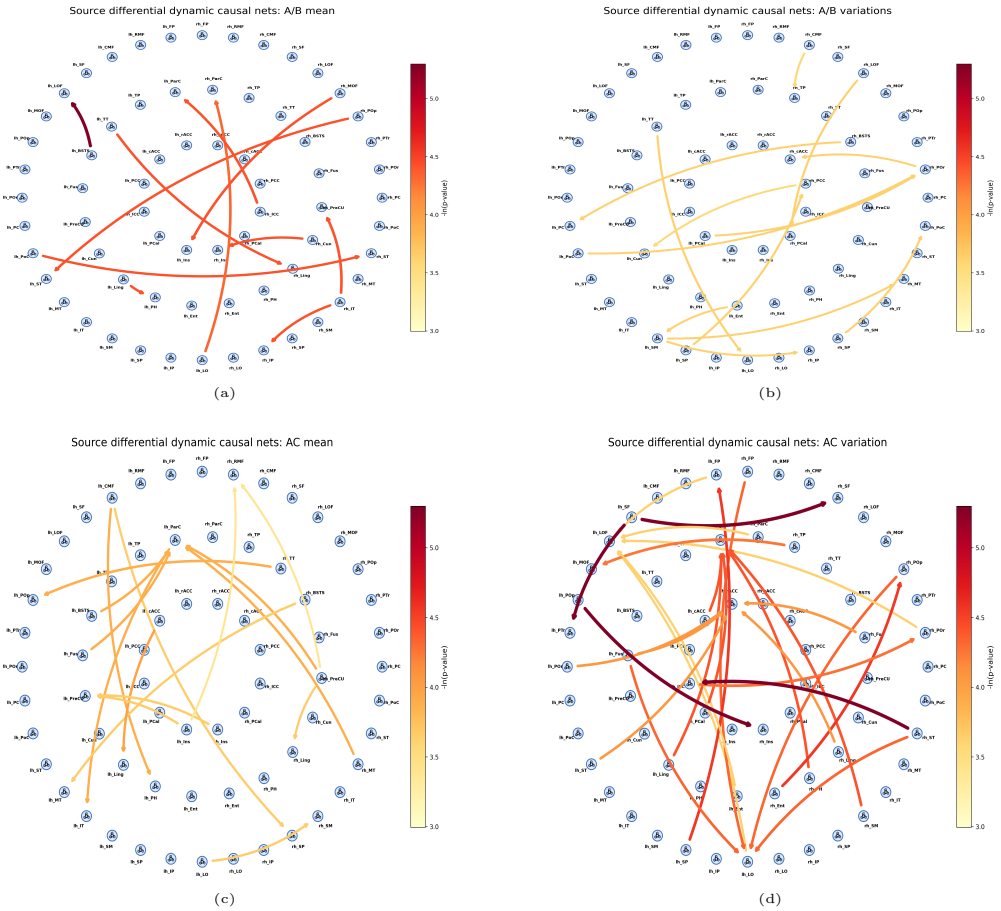


Figure 7. Source differential dynamic causal nets for parameters after the corrections for multiple testing and for multiple-input effects. (a) and (b): B_A mean and variation. (c) and (d): C_A mean and variation. The colour bars are in the scale of $-\ln$(p-value), where p-value is BH-adjusted.

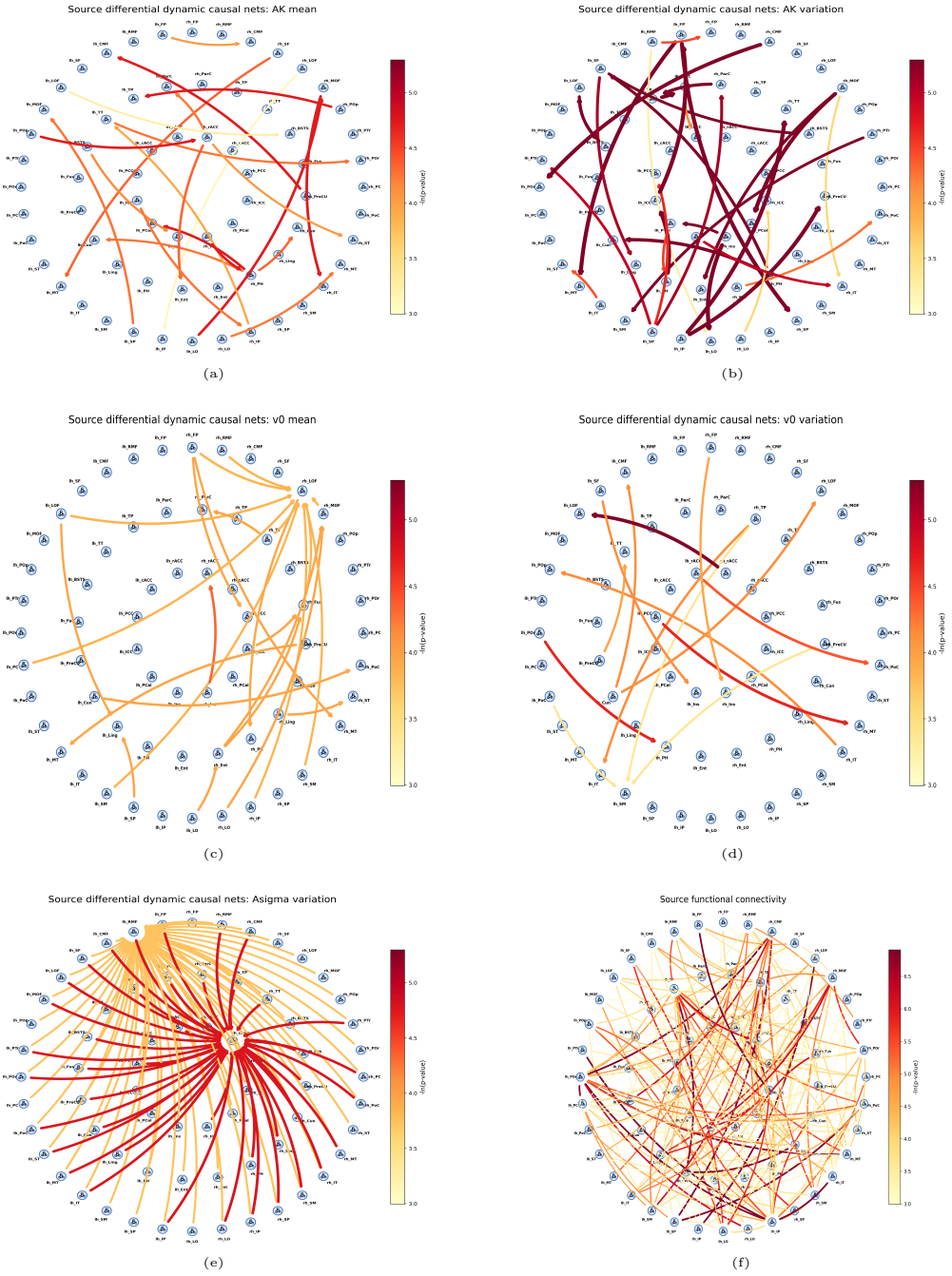


Figure 8. Source differential dynamic causal nets for parameters after the corrections for multiple testing and for multiple-input effects. (a) and (b): K_A mean and variation. (c) and (d): ν_0 mean and variation. (e): σ_A variation. (f): Source differential causal nets using the maximum lagged-correlation-based functional connectivity between the case and control groups at the level 5%. However, these edges in the nets were not significant at the level of 5% after the adjustments for multiple testing. The colour bars are in the scale of $-\ln(\text{p-value})$, where p-value is BH-adjusted.

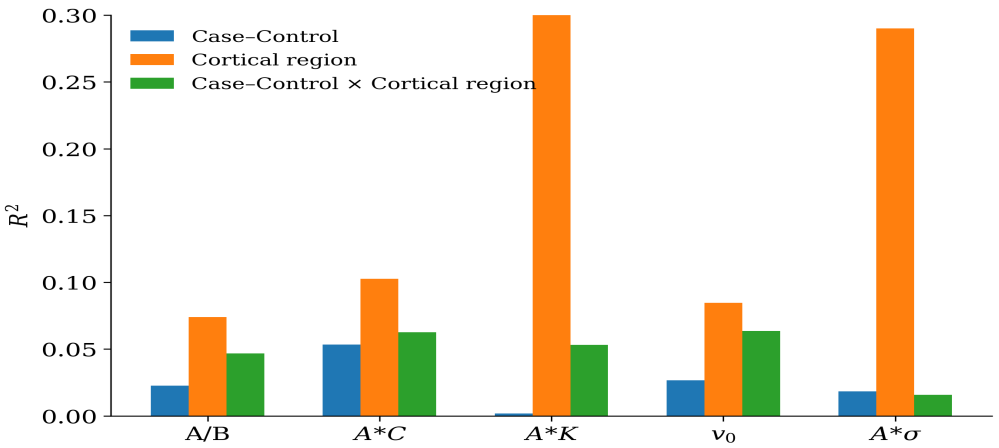


Figure 9. The percentages of uncertainty in the estimated source parameters which was explained by the NccDCM-based mixed-effects models, R^2. See Anderson (2017) for the definition of R^2. Here, $B_A = A/B, C_A = AC, K_A = AK$ and $\sigma_A = A\sigma$.

- In the ν_0 sensor differential dynamic causal net, the significant changes of input-output coefficients were identified in channel pairs (FP2F4,CZPZ) and (P3O1, T6O2), related to the connections between the frontal, temporal and posterior regions.

Source level analysis. We fitted a NccDCM to the estimated source time series for each subject in the preictal period and the ictal period respectively, followed by the pair-matched post Wilcoxon signed rank test for the period differences. Unlike the sensor-level analysis, source-level analysis revealed more significant changes in parameters, even in B_A. The results, displayed in Figure 11 below can be highlighted as follows.

- The B_A differential dynamic causal net demonstrates a state-dependent alteration of the input-output network topology during epileptic discharges, particularly involving mesial temporal, posterior cingulate, precuneus and superior parietal regions. The results were partially confirmed by previous EEG/MEG functional connectivity studies (Kumar et al., 2023; Sun et al., 2023; Liu et al., 2024). In addition, resting-state MEG/fMRI and structural-connectome studies demonstrate that several frontal and temporal regions—including fusiform, entorhinal, inferior temporal, transverse temporal, pars opercularis/orbitalis and anterior cingulate cortices—form a part of altered epileptic network (Liu et al., 2024; Schwarzer et al., 2026). Thus, these regions provide biologically plausible candidate nodes for seizure-state–dependent changes.

- The σ_A differential dynamic causal net shows that a few cortical regions are of high degree centrality, including the insula (lh-Insula and rh-Insula), anterior cingulate (rh-Caudalanteriorcingulate), posterior cingulate (lh-posteriorcingulate, rh-posteriorcingulate) and isthmus cingulate (lh-isthmuscingulate). Here, degree centrality defines the number of node's first-order neighbors in the network. This is consistent with the recent finding in a cortical thickness study (Ogren et al., 2018) that increased cortical thickness involving the insula, anterior cingulate, posterior cingulate and isthmus cingulate, as well as the pericalcarine cortex. Singh et al. (2020) using stereo-EEG found heterogeneous but focal seizure-onset

patterns within the insula. The anterior cingulate cortex was identified as a potential seizure-onset and epileptogenic region in focal epilepsy (Yang et al., 2023).

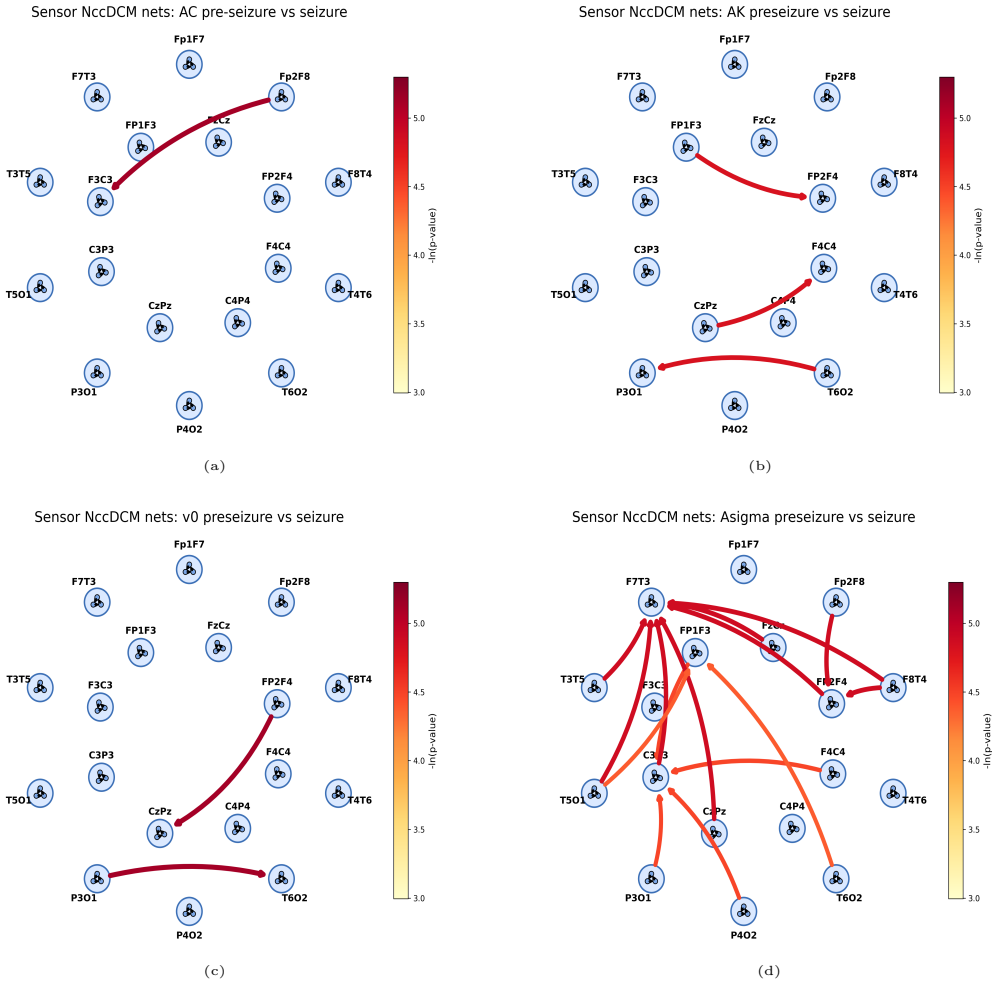


Figure 10. Sensor differential dynamic causal nets of parameters for pre-seizure/interictal v.s. seizure/ictal. (a) C_A , (b) K_A, (c) ν_0 and (d) σ_A. The colour bars are in the scale of $-\ln$(p-value), where p-value is BH-adjusted.

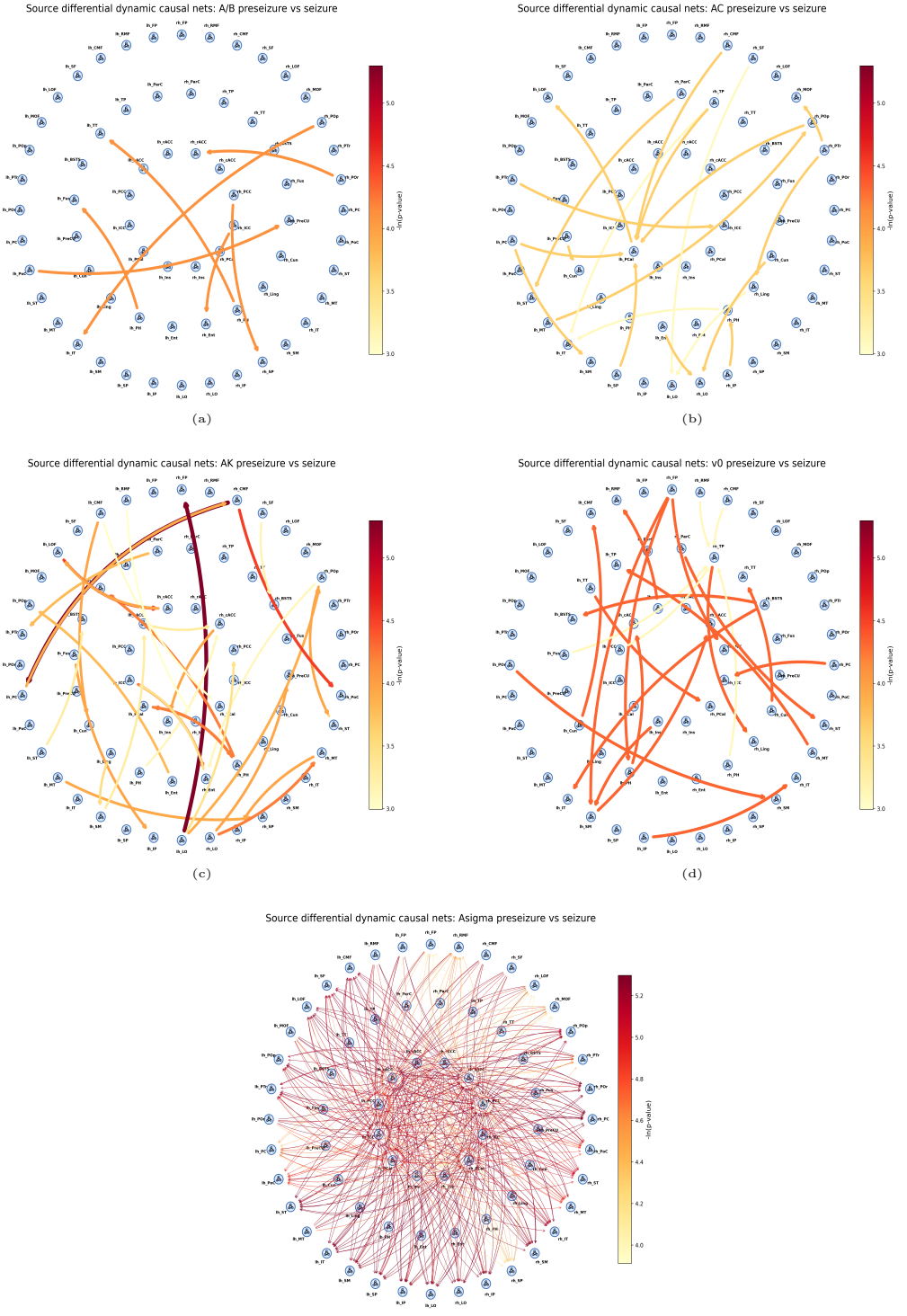
3.4 Simulation studies

3.4.1 Estimating a single node

Consider the following conditionally coupled Jansen-Rit model:

$$\begin{aligned} dx_0(t) &= x_3(t)dt, \quad dx_1(t) = x_4(t)dt, \quad dx_2(t) = x_5(t)dt, \\ dx_3(t) &= \left[AaS\left(x_1(t) - x_2(t)\right) - 2ax_3(t) - a^2x_0(t)\right]dt, \\ dx_4(t) &= \left[Aa\left\{C_2S\left(C_1x_0(t)\right) + KS\left(y_2(t)\right)\right\} - 2ax_4(t) - a^2x_1(t)\right]dt + Aadw(t), \\ dx_5(t) &= \left[BbC_4S\left(C_3x_0(t)\right) - 2bx_5(t) - b^2x_2(t)\right]dt, \end{aligned} \tag{8}$$

with $C_1 = C$, $C_2 = 0.8C, C_3 = 0.25C = C_4$, state vector $\boldsymbol{x}(t) = (x_0(t), x_1(t), x_2(t), x_3(t), x_4(t), x_5(t))$ and initial value



(a) (b) (c) (d) (e)

Figure 11. Source differential causal nets of parameters for pre-seizure/interictal v.s. seizure/ictal. (a) B_A, (b) C_A , (c) K_A, (d) ν_0 and (e) σ_A. The colour bars are in the scale of $-\ln$(p-value), where p-value is BH-adjusted.

$\boldsymbol{x}_0 = (x_0(0), ..., x_5(0))^T \in \mathbb{R}^6$ and the input $y_2(t)$ from the external channel. We use the Strang splitting method (see S12, the Online Supplementary Material) to generate the samples of $\boldsymbol{x}(t)$ from the above model. See Milstein and Tretyakov (2003); Leimkuhler et al. (2016) for more details. We can convert the above model to a canonical Jansen-Rit model/ccDCM by reparametrisation. As ccDCMs play a key role in shaping and influencing the structure and behavior of the NccDCM, we first showed the bifurcation behavior of ccDCM in S12, the Online Supplementary Material. Then, in the following, we evaluated the accuracy of the proposed estimation method. We measured the accuracy of the proposed model estimation by absolute relative error (ARE):

$$\text{ARE}(\widehat{\theta}) = |\widehat{\theta} - \theta^{(0)}|/|\theta^{(0)}|$$

with $\theta^{(0)} = (B_A, C_A, K_A, \nu_0, \sigma_A)$ as the ground truth. We calculated the median and the normalized median absolute deviation (NMAD) defined as $1.48 \times$ MAD of AREs.

We assessed the ARE of the proposed estimation for the ccDCM model (8) in the following settings :

Setting I. $A = 3.25$, $B = 22$, $C = 135$, $K = 50$, $v_0 = 6$ and $\sigma = 10$.

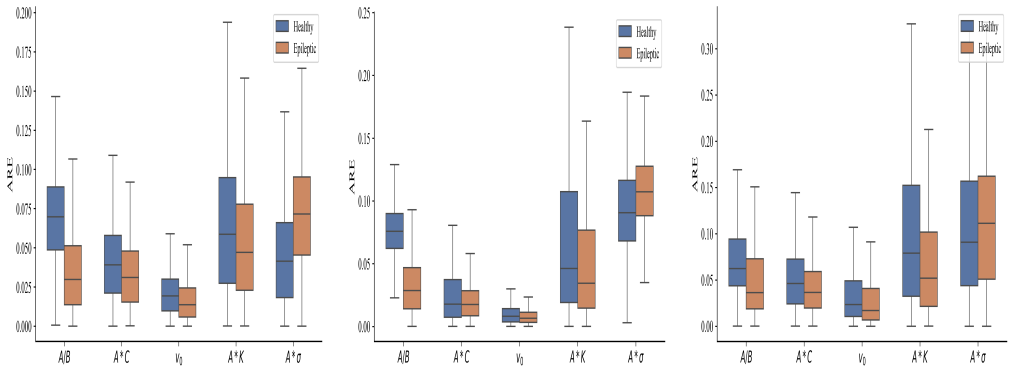
Setting II. $A = 4$, $B = 14$, and the remaining parameters are kept the same as in Setting I.

Setting III. $A \sim U(3, 6)$, $B \sim U(17.6, 30)$, $C \sim U(108, 160)$, $K \sim U(0, 50)$, $v_0 \sim U(3.12, 9)$ and $\sigma \sim U(5, 10)$, where $U(a, b)$ is a uniform distribution over the interval $[a, b]$.

In these settings, input time series $y_2(t)$ were collected from epileptic patients (P1, P3, and P5 to P10) and healthy subjects (H1 to H10) in the channels FP2F4, F4C4, C4P4, P4O2, FP1F3, F3C3, C3P3, P3O1, FP2F8, F8T4, T4T6, T6O2, FP1F7, F7T3, T3T5, T5O1, FZCZ, and CZPZ respectively. There were 10×18 time series from healthy subjects and 8×18 time series from epileptic subjects. Each of these time series were made of 6 segments, with the duration of 20 seconds each. We set the simulation step size $h^{\text{sim}} = 1/256$ and the block length of 2 (i.e., two time instances) to ensure that each time block was short enough so that the Chen–Fliess series approximation was accurate enough. For each input time series segment, we generated an output time series segment from the model (8). Totally, we simulated $M_H = 1080$ healthy input-output time series segment pairs and $M_P = 864$ epileptic input-output time series segment pairs. For each input-output time series segment pair, we fitted it by ccDCM, resulting in estimated parameter values of $(B_A, C_A, K_A, \nu_0, \sigma_A)$. Compared them to the ground truth, we then evaluated the median and NMADs of AREs for simulated healthy segment pairs and for simulated epileptic segment pairs respectively. The results, displayed by boxplots in Figure 12 and by Table 2, showed that the proposed estimation is promising.

Table 2. Median ARE values with corresponding NMADs (in parentheses) for ccDCM estimation. The left and right blocks present results obtained using EEG time series from epileptic and healthy subjects as external inputs, respectively.

Epileptic				Healthy			
Parameter	**Setting I**	**Setting II**	**Setting III**	**Parameter**	**Setting I**	**Setting II**	**Setting III**
B_A	0.03 (0.03)	0.03 (0.02)	0.04 (0.03)	B_A	0.07 (0.03)	0.08 (0.02)	0.06 (0.03)
C_A	0.03 (0.02)	0.02 (0.01)	0.04(0.03)	C_A	0.04 (0.03)	0.02 (0.02)	0.05 (0.04)
ν_0	0.01 (0.01)	0.01 (0.01)	0.02 (0.02)	ν_0	0.02 (0.01)	0.01 (0.01)	0.02 (0.02)
K_A	0.05 (0.04)	0.03 (0.04)	0.05 (0.05)	K_A	0.06 (0.05)	0.05 (0.05)	0.08 (0.08)
σ_A	0.07 (0.04)	0.11 (0.03)	0.11 (0.08)	σ_A	0.04 (0.03)	0.09 (0.04)	0.09 (0.08)



Setting I Setting II Setting III

Figure 12. Box-plots of ARE values of the estimated $B_A = A/B$, $C_A = AC$, v_0, $K_A = AK$ and $\sigma_A = A\sigma$ compared to the ground truths in Settings I, II, III.

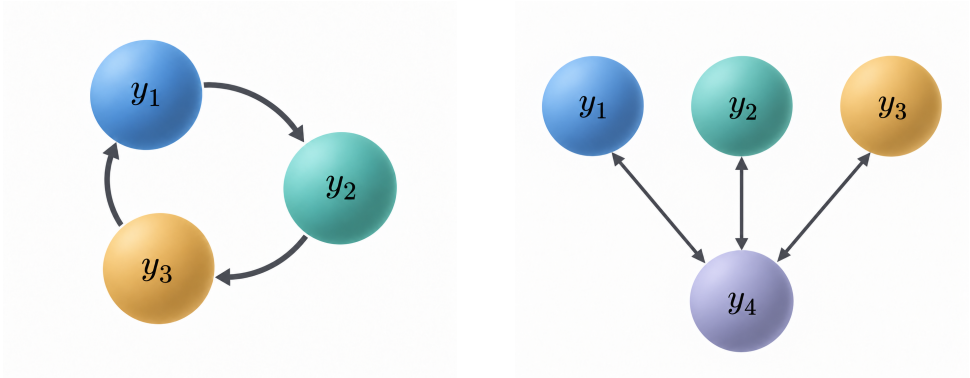
3.4.2 Joint estimation of multiple nodes.

In the following, we assessed the ARE of the proposed estimation for multiple nodes, where each node stands for a Jansen-Rit neural mass model. Note that the proposed estimation of NccDCM depends on the methods of solving the Jansen-Rit equations. In this paper, we proposed using Chen-Fliess method, instead of standard numerical solvers such as the RK and EM schemes to solve the Jansen-Rit differential equations. We consider two distinct multi-node configurations: one with unidirectional coupling and the other with bidirectional coupling, as illustrated in Figure 13. The detailed mathematical forms can be found in S13, the Online Supplementary Material.

For unidirectional coupling setting, we assessed the proposed method against RK and EM standard solver-based methods. Note that the RK approximation for the stochastic Jansen–Rit model is computed using the function `stratSRS2` from the Python library `sdeint`. The parameters in unidirectional coupling setting were set as follows: $A^{(1)} = 3.25$, $B_A^{(1)} = 3.25/22$, $C_A^{(1)} = 3.25 \times 135$, $K_A^{(1|3)} = 3.25 \times 50$, $\nu_0^{(1)} = 6$, $\sigma_A^{(1)} = 3.25 \times 10$; $A^{(2)} = 4$, $B_A^{(2)} = 4/14$, $C_A^{(2)} = 4 \times 135$, $K^{(2|1)} = 4 \times 70$, $\nu_0^{(2)} = 4$, $\sigma_A^{(2)} = 4 \times 10$; $A^{(3)} = 3$, $B^{(3)} = 3/15$, $C_A^{(3)} = 3 \times 135$, $K^{(3|2)} = 3 \times 50$, $\nu_0^{(3)} = 5$, $\sigma_A^{(3)} = 3 \times 10$, where node 1 receives input $y^{(3)}(t)$ from node 3, node 2 receives input $y^{(1)}(t)$ from node 1, and node 3 receives input $y^{(2)}(t)$ from node 2. The simulation time interval was $[0, T] = [0, 20]$ seconds with the sampling frequency to 256 Hz. To reduce the influence of the initial conditions on the loss function, the first second simulated time series was discarded and the remaining time series was used to calculate the loss function. To further examine the robustness of the proposed method with respect to the choice of initial conditions, the following three scenarios were considered: (i) zero initial conditions $(0, 0, 0, 0, 0, 0)$; (ii) constant initial conditions $(0, 0.5, 1, 1.5/A^{(i)}, 2, 2.5)$; and (iii) random initial conditions drawn from normal distributions, namely, $x_{A0}^{(i)}(0) \sim N(0, 1/A^{(i)}), x_1^{(i)}(0) \sim N(0, 1), x_2^{(i)}(0) \sim N(0, 1), x_{A3}^{(i)}(0) \sim N(0, 1/A^{(i)}), x_4^{(i)}(0) \sim N(0, 1), x_5^{(i)}(0) \sim N(0, 1)$. We repeatedly simulated the triple time series $(y^{(i)}(t) : i = 1, 2, 3), t \in [0, T]$ 100 times from the above model. The AREs of estimated parameters in NccDCM were calculated for the Chen-Fliess-, RK- and EM-based schemes respectively. The simulation results were summarised in Table 3 and Table S16, the Online Supplementary Material. The results demonstrated that the Chen-Fliess method outperformed the RK- and EM- schemes in terms of ARE. The results in Table S16, the Online Supplementary Material, also suggested that the Chen-Fliess method was robust against different initial state

conditions. Furthermore, Table 4 indicated that the Chen-Fliess method had much lower CPU time costs than the RK-based method when we ran these procedures in the High-Performance Computing Service provided by the University of Kent. The EM-scheme had lower CPU time costs than the Chen-Fliess (the details are omitted).

For the bidirectional coupling configuration, the simulation results, displayed in Tables S17 and S18, the Online Supplementary Material, suggested that the multiple-input based parameter estimation outperformed the pairwise parameter estimation in terms of ARE.



(a) Unidirectional coupling (b) Bidirectional coupling

Figure 13. Schematic illustrations of two multi-node configurations. y_i is the output of node i while arrows show the direction of coupling between these nodes. (a) Unidirectional coupling configuration. (b) Bidirectional coupling configuration.

4 Discussion and conclusions

Enabling data-driven composite network modelling at whole brain scales. The question as to how large-scale dynamic causal patterns emerge from underlying neuronal interactions is one that has attracted much interest in neurological disease studies (Friston et al., 2019; Forrester et al., 2020; Pereira et al., 2021; Forrester et al., 2024; Singh et al., 2025) and in statistics (Pokern et al., 2009; Ditlevsen et al., 2025). The existing imaging studies often focused on a single subject study, either with models for a small number of cortical columns or with models in a large scale but involving only excitatory/inhibitory neuron populations or with models in a large scale but for homogeneous brain regions, where all nodes in the network models are identical. In this paper, to relax the above limitations, we have proposed a heterogeneous composite network model, NccDCM, for groups of EEG scans respectively. In the NccDCM, each node represents a local Jansen-Rit neural mass model for describing synaptic interactions in excitatory/inhibitory and pyramidal neuron populations, which consists of six-dimensional stochastic differential equations, while directed edges stand for coupling/transmissions between these local neuron populations across different channels. To account for individual subject-variability, we have introduced a mixed-effects model for network parameters. We have developed the Chen-Fliess expansion-based evolutionary optimisation algorithm for estimating NccDCMs. For ease of exposition, we have adopted a pairwise screening strategy to estimate edges in the proposed composite network. To identify dynamic causal nets that differs cases from controls, we have carried out a post-analysis of variance for estimated parameters in cases and controls. In the theoretical study, we have revealed that the excitation and inhibition parameters are not structurally identifiable although

Table 3. Median and NMAD (in parenthesis) of ARE values for the EM-, RK- and CF-based methods with zero initial state values. EM, RK and CF stand for Euler-Maruyama, Runge-Kutta and Chen-Fliess. The bold face entries are the minimum values in their corresponding rows.

Parameter	Node	EM-based	RK-based	CF-based
B_A	1	0.40 (0.18)	0.55 (0.17)	**0.05** (0.04)
	2	0.80 (0.85)	0.77 (0.90)	**0.02** (0.02)
	3	0.18 (0.19)	0.16 (0.17)	**0.03** (0.02)
C_A	1	0.39 (0.09)	0.42 (0.13)	**0.03** (0.02)
	2	0.97 (0.04)	0.96 (0.05)	**0.04** (0.03)
	3	0.34 (0.08)	0.34 (0.16)	**0.03** (0.03)
v_0	1	0.09 (0.06)	0.08 (0.07)	**0.03** (0.02)
	2	0.20 (0.19)	0.26 (0.20)	**0.02** (0.02)
	3	0.20 (0.10)	0.19 (0.09)	**0.02** (0.02)
K_A	1	0.34 (0.18)	0.55 (0.21)	**0.09** (0.07)
	2	0.32 (0.06)	0.31 (0.06)	**0.11** (0.12)
	3	0.10 (0.07)	0.11 (0.12)	**0.05** (0.04)
σ_A	1	0.84 (0.01)	0.83 (0.02)	**0.02** (0.02)
	2	0.86 (0.02)	0.85 (0.03)	**0.10** (0.04)
	3	0.82 (0.01)	0.82 (0.02)	**0.03** (0.03)

Table 4. Mean computation time and corresponding standard deviation (Std) for the Chen-Fliess (CF)-based and RK-based methods based on 200 replications.

Method	Mean Time (sec)	Std (sec)
CF-based	**570.507**	54.693
RK-based	3894.649	356.709

their ratio is locally structurally identifiable. A similar phenomenon has been found in conductance-based DCM by Hauke et al. (2025). This has given rise to the approach of reparametrisation of the original parameters that the new parameters, including the excitation/inhibition balance ratio, are locally structurally identifiable except the excitation parameter. We have assessed the absolute relative errors of the model estimation, demonstrating that the performance of the proposed estimates for the node and edge parameters is relatively accurate.

Detecting and quantifying differential epileptic network. Epilepsy is considered a network disease that affects the brain across multiple levels of spatial and temporal scales. The epileptic network, however, is not static but evolves in time which requires novel approaches for an in-depth characterisation (Bröhl et al., 2024). The proposed biophysics-informed network of neural mass models aims to detect and quantify such a network. In particular, we have pursued a computational approach to elucidating the difference of brain dynamic connectivity patterns called dynamic causal nets between cases and controls or between the preictal period and the ictal period. We first fitted a NccDCM to the EEG scan for each subject in the case and control groups respectively. Following that, a mixed-effects model-based post-analysis of variance has been conducted to contrast estimated case-control parameters. This has given rise to various differential causal nets that differ the case-networks from the control-networks. The same analysis has also been performed on the segments of time series in preictal and ictal periods. We have found similar differential causal nets that differ the preictal period from the ictal period. These nets are associated with epileptic seizures even after the Benjamini-Hochberg correction for multiple testing.

It is widely recognised that EEG scans are inherently noisy with weak signals due to the minuscule amplitude of brain-generated electrical activity recorded at the scalp and the susceptibility to various sources of contamination. Only the synchronous firing of millions of neurons produces a measurable signal at the scalp. Despite this, we have demonstrated the advantage of the NccDCM analysis over the conventional temporal correlation-based analysis in detecting links between regions. We have identified a list of dynamic causal nets to characterise the spatial-temporal varying epileptic network. These differential dynamic causal nets partially covered default mode network (DMN), a collection of distributed and interconnected brain regions. DMN is typically suppressed when the brain receives external stimuli; however, in the absence of attention to external stimuli, the DMN switches to internally focused thought processes. Therefore, the DMN activity can be observed in subject's resting states. It follows from Menon (2023) that the DMN nodes include posterior cingulate cortex (PCC) and retrosplenial cortex (RSC) in posterior medial parietal cortex; medial prefrontal cortex PFC (mPFC) with its dorsomedial (dmPFC) and ventromedial (vmPFC) subdivisions; anterior temporal cortex (ATC); middle temporal gyrus (MTG) in lateral temporal cortex; medial temporal lobe (MTL); and angular gyrus (AG) in lateral parietal cortex; and caudate nucleus (CAU). Activities in PCC and RSC are captured by electrodes FP2, P3, P4 and O2; AG is captured by P3, P4, T5 and T6; PFC is captured by FP1. FP2, Fz, F3 and F4; ATC is captured by T3 and T4; MTG is captured by T5 and T6; MTL is captured by F3, T4, T5, and T6. EEG channels do not directly cover subcortical structures like the thalamus (THAL) and caudate nucleus (CAU) because these deep brain regions do not produce sufficient electrical signals that reach the scalp surface to be localised to a single channel. In Figures 4 and 5, the identified dynamic causal nets differ cases from controls in the means and variations of the estimated NccDCM parameters. These nets imply epilepsy-predisposed changes in the DMNs: (a) Edge changes in the excitatory/inhibitory synaptic ratio: AG/PCC points to PFC. (b) Many edge changes

in the local structure parameters C_A. (c) Changes in the transmission parameter K_A: ATC points to PFC; AG points to PFC. (d) Edge changes in ν_0: AG points to PFC; ATC points to PFC. Figure 10, compared the ictal period to the preictal period, yields many dynamic causal nets that have been related to the estimated region-transmission parameters. This finding implies that the changes in the region-transmission parameters can be used for chracterising onset of seizures.

Our findings have provided a few epileptic network configurations than the DMN, highlighting the contribution of dysfunctions in the DMN to epileptic activities and cognitive behaviors in patients e.g., (e.g., Jiang et al., 2022). As epileptic seizures are associated with abnormal neuronal hyperexcitability, we hypothesize that alterations in the excitation–inhibition balance, local connectivity, or interregional connectivity within the patients' default mode network (DMN) are associated with epilepsy.

Source level analysis. Although the cortical regions identified in the B_A differential causal net have previously been implicated in epilepsy-related structural and functional abnormalities. Epilepsy is increasingly conceptualized as a disorder involving distributed structural and functional brain networks that can extend beyond the putative seizure-onset region (Whelan et al., 2018). However, the patients have heterogeneous and differently lateralised epilepsies, so group-level left- or right-hemisphere findings need to align the side of onset.

The regions identified in the present analysis included the left banks of the superior temporal sulcus (bankssts), left lateral orbitofrontal cortex, bilateral lingual cortex, left parahippocampal cortex, left postcentral cortex, bilateral superior temporal cortex, left transverse temporal cortex, right cuneus, bilateral insula, right inferior temporal cortex, right inferior parietal cortex, right precuneus, right isthmus cingulate cortex, bilateral paracentral cortex, right medial orbitofrontal cortex, right pars opercularis, and left lateral occipital cortex.

The temporal and limbic regions identified in the present analysis are also consistent with the network-level alterations frequently reported in temporal lobe epilepsy. In particular, the parahippocampal and superior temporal regions form part of distributed temporal networks that may show structural and functional abnormalities in epilepsy. The insular cortex warrants particular consideration. The insula has extensive anatomical and functional connections with frontal, temporal, parietal, and subcortical regions and can participate in both seizure generation and seizure propagation (Solanki et al., 2023). Insular seizures can also produce clinical manifestations that resemble seizures originating from temporal, frontal, or parietal regions. Consequently, increased connectivity involving the insula may reflect its role as part of a distributed epileptic network rather than necessarily indicating an isolated insular seizure-onset zone (Ryvlin and Nguyen, 2021; Solanki et al., 2023).

Overall, the presence of these regions among the areas showing altered directed connectivity is broadly consistent with previous evidence that epilepsy involves distributed cortico-cortical and cortico-subcortical networks. Nevertheless, the present connectivity results alone cannot establish that any individual region is epileptogenic. Rather, they indicate that these regions participate in the pattern of directed functional interactions observed in the studied population. This distinction is particularly important when interpreting regions showing high in-degree or high out-degree. A high in-degree indicates that a region receives directed connections from many other regions, whereas a high out-degree indicates that a region sends directed connections to many other regions. These network measures therefore describe the organization of directed connectivity and should not be interpreted directly as measures of seizure onset or epileptogenicity.

Limitations. We view our present model as having the following limitations: 1) algorithms for estimating the NccDCM and 2) the omission of subjects' demographic information. We have developed a marginal screening approach to estimating the node-wise connection parameters. Marginal screening is a method used to estimate the coefficients of multiple external node inputs by identifying the most significant coefficients for each node. This approach is particularly useful when a large number of nodes involved. The method involves constructing a model that includes all external nodes and then selecting the most significant nodes to include in the final model. Our simulation studies have demonstrated that this method helps to avoid overfitting and ensures that the model is generalisable to new data. However, the method is at the cost of estimation efficiency. The penalisation regression or forward/backward variable selection has been explored in another paper. The second limitation potentially arise from the neglect of subjects' demographic information such as age and gender. However, the numbers of sujects in the case and control groups are relatively small. A further work is needed to take age and gender as additional predictors in each ccDCM for increased sizes of case-control gropus.

Conclusions and future work. In summary, the proposed NccDCM is a novel and useful tool for detecting disease-predisposed effective connectivity in case-control study using EEG data. We have shown that the NccDCM can be used to characterise epileptic brain network at the whole brain scale in both sensor and source levels. The proposed method also demonstrates promising performance on synthetic data. This work can be extended to other neural mass models in Friston et al. (2019).

Appendix A

Proposition 1. The canonical Jansen-Rit model (2) is locally structurally identifiable.

Proof: Note that the canonical Jansen-Rit model has non-rational vector field functions while the existing software SIAN for structural identifiability analysis is only applicable to neural mass differential equations with rational vector field functions. In order to use the software SIAN to analyse the identifiability of the canonical Jansen-Rit model with zero noises, we first bring the canonical Jansen-Rit model to the form with rational vector field functions described in Hong et al. (2019) by introducing new states $x_i(t), i = A3, 4, 5, 6, 7, 8$:

$$\begin{aligned}
\dot{x}_{A0} &= x_{A3}, \quad \dot{x}_1(t) = x_4(t), \quad \dot{x}_2(t) = x_5(t), \\
\dot{x}_{A3}(t) &= ax_6(t) - 2ax_{A3}(t) - a^2x_{A0}(t), \\
\dot{x}_4(t) &= a[P(t) + 0.8C_Ax_7(t)] - 2ax_4(t) - a^2x_1(t), \\
\dot{x}_5(t) &= B_A^{-1}b0.25C_Ax_8(t) - 2bx_5(t) - b^2x_2(t), \qquad (9) \\
\dot{x}_6(t) &= rx_6(t)(1 - x_6(t))(x_4(t) - x_5(t)), \\
\dot{x}_7(t) &= rx_7(t)(1 - x_7(t))C_Ax_{A3}(t), \\
\dot{x}_8(t) &= rx_8(t)(1 - x_8(t))0.25C_Ax_{A3}(t),
\end{aligned}$$

(10)

where

$$\begin{aligned}
x_6(t) &= 1/(1 + \exp(r(\nu_0 - x_1(t) + x_2(t))), \quad x_7(t) = 1/(1 + \exp(r(\nu_0 - C_Ax_{A0}(t))), \\
x_8(t) &= 1/(1 + \exp(r(\nu_0 - 0.25C_Ax_{A0}(t))). \quad P(t) = \sum_{j \neq k} K_A^{(k|j)} y_j(t)
\end{aligned}$$

with the output $(y_j(t) : j \in I_k)$. The SIAN analysis shows that the parameters $B_A, C_A, P(0)$, $x_{A0}(0), x_1(0), x_2(0), x_{A3}(0)$, $x_i(t)$, $i = 4, 5, 6, 7, 8$ in equations (9) are locally structurally identifiable, but has not identified any globally structurally identifiable parameters.

It follows from the definition that

$$\begin{aligned} \nu_0 &= r^{-1} \log\left(\frac{1 - x_7(0)}{x_7(0)}\right), \\ r &= \frac{1}{\nu_0 - 0.25 C_A x_{A0}(0)} \log\left(\frac{1 - x_8(0)}{x_8(0)}\right) \end{aligned}$$

which imply that given the external input $z(t)$, parameters ν_0 and r are also locally structurally identifiable. Note that $P(t)$ is identifiable as $P(0)$ is identifiable. Given time instances $t_1 < \cdots < t_m$ with m being larger than the size $|I_k|$ of I_k, we have

$$\begin{pmatrix} P(t_1) \\ \vdots \\ P(t_m) \end{pmatrix} = \begin{pmatrix} y_1(t_1) & \cdots & y_{|I_k|}(t_1) \\ \vdots & & \vdots \\ y_1(t_m) & \cdots & y_{|I_k|}(t_m) \end{pmatrix} \begin{pmatrix} K^{(k|1)} \\ \vdots \\ K^{(k||I_k)|} \end{pmatrix}$$

which yields unique values $K^{(k|j)}, j \in I_k$ under the condition (**C0**). The proof is completed.

Appendix B

Consider the following set of differential system of the form

$$\dot{z}(t) = g_0(z(t)) + \sum_{i=1}^{m} g_i(z(t)) u_i(t), \quad z(0) = z_0; \quad y(t) = h(z(t)), \tag{11}$$

where each g_i is an analytic vector field expressed in some neighborhood of z_0, and the h is an analytic output function. Assume that (11) has a well defined solution $z(t)$, $t \in [t_0, T + t_0]$ for any given input $u_i(t)$, and $y(t) = h(z(t))$, $t \in [t_0, T + t_0]$.

Lemma The solution to equation (11) can be expressed as a Chen-Fliess series of the form

$$\begin{aligned} y(t) = & \sum_{1 \le i_1, \ldots, i_k \le m, k \ge 0} \left(\int_0^t \int_0^{\tau_k} \cdots \int_0^{\tau_2} u_{i_k}(\tau_k) \cdots u_{i_1}(\tau_1) d\tau_1 \cdots d\tau_{k-1} d\tau_k \right) \\ & \times \left(L_{g_{i_1}} \circ \cdots \circ L_{g_{i_k}} h(z_0) \right), \end{aligned} \tag{12}$$

where $L_{g_{i_1}} \circ \cdots \circ L_{g_{i_k}} h(z_0)$ is a Lie-derivative in the direction of the vector field. See Lyons (1998).

Chen-Fliess expansion of states in model (8)). For a small increment h and $t \in [t_0, t_0 + h]$, truncating the Chen-Fliess expansion of $x_0(t)$ and $x_2(t)$ at the 3rd

order, we have

$$\begin{aligned} x_{A0}(t) &= x_{A0}(t_0) + x_{A3}(t_0)(t-t_0) \\ &\quad + \left(\frac{\nu_{\max} a}{1+e^{r(v_0 - y(t_0))}} - a^2 x_{A0}(t_0) - 2a x_{A3}(t_0) \right) \frac{(t-t_0)^2}{2} \\ &\quad + \left(\frac{r\nu_{\max} a e^{r(v_0-y(t_0))} \dot{y}(t_0)}{\left(1+e^{r(v_0-y(t_0))}\right)^2} - a^2 x_{A3}(t_0) \right. \\ &\quad \left. -2a \left(\frac{\nu_{\max} a}{1+e^{r(v_0-y(t_0)))}} - a^2 x_{A0}(t_0) - 2a x_{A3}(t_0) \right) \right) \frac{(t-t_0)^3}{3!} \\ &\quad + O_p((t-t_0)^4), \end{aligned} \tag{13}$$

$$\begin{aligned} x_2(t) &= x_2(t_0) + x_5(t_0)(t-t_0) \\ &\quad + \left(\frac{\nu_{\max} B_A^{-1} b 0.25 C_A}{1+e^{r(v_0 - C_A x_{A0}(t_0))}} - 2b x_5(t_0) - b^2 x_2(t_0) \right) \frac{(t-t_0)^2}{2} \\ &\quad + \left(\frac{r\nu_{\max} B_A^{-1} b (0.25 C_A)^2 x_{A3}(t_0) e^{r(v_0 - 0.25 C_A x_{A0}(t_0))}}{(1+e^{r(v_0-0.25 C_A x_{A0}(t_0))})^2} - b^2 x_5(t_0) \right. \\ &\quad \left. -2b \left(\frac{\nu_{\max} B_A^{-1} b 0.25 C_A}{1+e^{r(v_0-0.25 C_A x_{A0}(t_0))}} - 2b x_5(t_0) - b^2 x_2(t_0) \right) \right) \frac{(t-t_0)^3}{3!} \\ &\quad + O_p((t-t_0)^4). \end{aligned} \tag{14}$$

Similarly, given the input $y^{(j)}(t)$ from channel j, truncating the Chen-Fliess expansion of $x^(t)$ at the 2nd order, we have

$$\begin{aligned} x_1(t) &= x_1(t_0) + x_4(t_0)(t-t_0) \\ &\quad + \left(\frac{\nu_{\max} a 0.8 C_A}{1+e^{r(v_0 - C_A x_{A0}(t_0))}} + \frac{\nu_{\max} a K_A}{1+e^{r(v_0 - y^{(j)}(t_0))}} - a^2 x_1(t_0) - 2a x_4(t_0) \right) \frac{(t-t_0)^2}{2} \\ &\quad + \sigma_A a \int_{t_0}^{t} (t-s) d\tilde{w}_s + o_p((t-t_0)^3), \end{aligned} \tag{15}$$

It follows from equations (14) and (15) that the truncated Chen-Fliess expansion of $y(t)$ at the 2nd order can be written as

$$y(t) \approx g(t | x_{-\{3\}}(t_0), y(t_0), \dot{y}(t_0), y^{(j)}(t_0), \boldsymbol{\theta} + \sigma_A a \int_{t_0}^{t} (t-s) d\tilde{w}_s, \tag{16}$$

where

$$\begin{aligned} & g(t | x_{-\{3\}}(t_0), y(t_0), \dot{y}(t_0), y^{(j)}(t_0), \boldsymbol{\theta}) \\ &= y(t_0) + \dot{y}(t_0)(t-t_0) \\ &\quad + \left(\frac{\nu_{\max} a 0.8 C_A}{1+e^{r(v_0 - C_A x_{A0}(t_0))}} + \frac{\nu_{\max} a K_A}{1+e^{r(v_0 - y^{(j)}(t_0))}} - a^2 x_1(t_0) - 2a x_4(t_0) \right) \frac{(t-t_0)^2}{2} \\ &\quad - \left(\frac{\nu_{\max} B_A^{-1} b 0.25 C_A}{1+e^{r(v_0 - 0.25 C_A x_{A0}(t_0))}} - 2b x_5(t_0) - b^2 x_2(t_0) \right) \frac{(t-t_0)^2}{2}. \end{aligned}$$

It is clear that g function depends on the initial values of states $x_{-\{A3\}}(t_0)$.

Let

$$\begin{aligned} L_A(t_0) &= \frac{\nu_{\max} a}{1+e^{r(v_0 - y(t_0))}} - a^2 x_{A0}(t_0) - 2a x_{A3}(t_0) \\ L_B(t_0) &= \frac{\nu_{\max} B_A^{-1} b 0.25 C_A}{1+e^{r(v_0 - 0.25 C_A x_{A0}(t_0))}} - 2b x_5(t_0) - b^2 x_2(t_0). \end{aligned}$$

The 3rd-order Chen-Fliess series of x_{A0} and x_2 are

$$\begin{aligned} & g_0(t|x_{\{A0,A3\}}(t_0), y(t_0), \dot{y}(t_0)) \\ = \; & x_{A0}(t_0) + x_{A3}(t_0)(t-t_0) + L_A(t_0)\frac{(t-t_0)^2}{2} \\ + \; & \left(\frac{r\nu_{\max} a e^{r(v_0 - y(t_0))}\dot{y}(t_0)}{\left(1+e^{r(v_0-y(t_0))}\right)^2} - a^2 x_{A3}(t_0) - 2aL_A(t_0) \right) \frac{(t-t_0)^3}{3!}, \end{aligned} \tag{17}$$

and

$$\begin{aligned} g_2(t|x_{-\{1,4\}}(t_0)) \; = \; & x_2(t_0) + x_5(t_0)(t-t_0) + L_B(t_0)\frac{(t-t_0)^2}{2} \\ & + \Bigg(\frac{r\nu_{\max} B_A^{-1} b 0.25 C_A 0.25 C_A x_{A3}(t_0) e^{r(v_0 - 0.25 C_A x_{A0}(t_0))}}{(1+e^{r(v_0-0.25C_A x_{A0}(t_0))})^2} \\ & - b^2 x_5(t_0) - 2bL_B(t_0) \Bigg) \frac{(t-t_0)^3}{3!}, \end{aligned} \tag{18}$$

respectively.

The 2nd-order Chen-Fliess series of x_3 and x_5 are

$$\begin{aligned} & g_3(t|x_{\{A0,A3\}}(t_0), y(t_0), \dot{y}(t_0)) = x_{A3}(t_0) + L_A(t_0)(t-t_0) \\ & + \left(\frac{r\nu_{\max} a e^{r(v_0 - y(t_0))}\dot{y}(t_0)}{\left(1+e^{r(v_0-y(t_0))}\right)^2} - a^2 x_{A3}(t_0) - 2aL_A(t_0) \right) \frac{(t-t_0)^2}{2}, \end{aligned} \tag{19}$$

and

$$\begin{aligned} g_5(t|x_{-\{1,4\}}(t_0)) \; = \; & x_5(t_0) + L_B(t_0)(t-t_0) \\ & + \Bigg(\frac{r\nu_{\max} B_A^{-1} b (0.25 C_A)^2 x_{A3}(t_0) e^{r(v_0 - 0.25 C_A x_{A0}(t_0))}}{(1+e^{r(v_0-0.25C_A x_{A0}(t_0))})^2} \\ & - b^2 x_5(t_0) - 2bL_B(t_0) \Bigg) \frac{(t-t_0)^2}{2}, \end{aligned} \tag{20}$$

respectively.

Author Contributions

K.Y. conceptualisation, methodology, coding, data analysis, writing & editing of original draft; G.G. conceptualisation, methodology, review & editing of original draft; J.Z. conceptualisation, methodology, coding, data analysis, writing and editing of original draft, supervision, funding acquisition and project administration.

Data and Code Availability

Codes to reproduce simulation results are in https://github.com/zhangjsib/NccDCM. The EEG data on epilepsy is publicly available in Fasil et al. (2021).

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Declaration of Competing Interests

K.Y., J.Z. and G.G. have no competing interests.

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Supplementary material

Supplementary material in a pdf file includes additional information as follows.

S1: A Lie symmetry analysis of the canonical Jansen-Rit model.

S2: An introduction on the optimisation algorithm JADE.

S3: Multiple permutation-based Wilcoxon signed-rank tests for significant NccDCM changes from the preictal period to the ictal period.

S4: Bootstrap bias correction.

S5: Lagged correlation-based functional connectivity.

S6: EEG electrode allocation on the scalp.

S7: Post-ANOVA tables.

S8: More results on subject-heterogeneity analysis.

S9: Bipolar EEG source localisation.

S10: Region of Interest Desikan-Killiany atlas.

S11: Bifurcation behavior of ccDCMs.

S12: Unidirectional circle network of three nodes.

S13: Investigate the more general case where one channel receives more than one inputs from other channels.

The pdf file of the Supplementary Material is available in https://github.com/zhangjsib/NccDCM.

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